

Lecture 3:

Vector Calculus



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**Numerical Methods for
Computational Science**

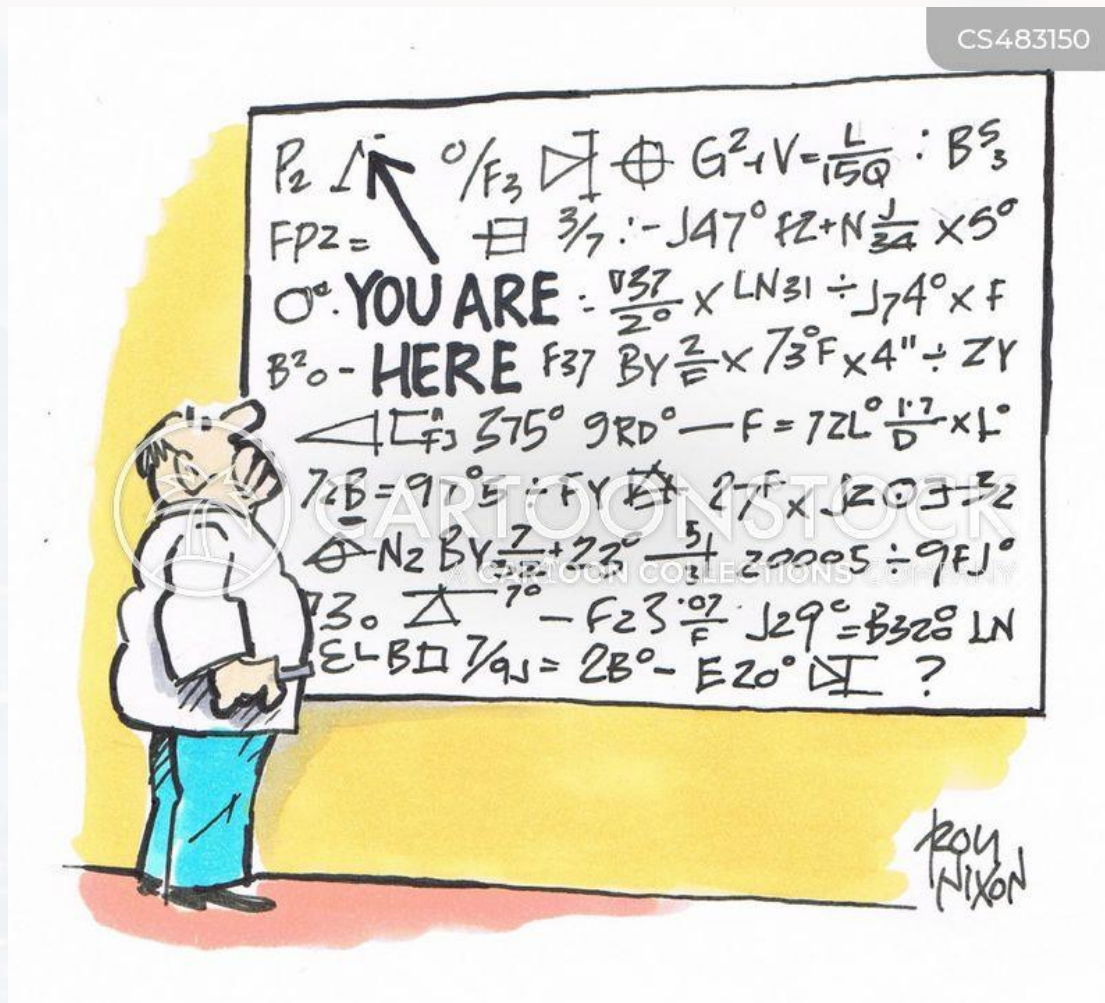
MSSE 273, 3 Units

Course Map

Week 1:	Introduction to Scientific Computing and Python Libraries
Week 2:	Linear Algebra Fundamentals
Week 3:	Vector Calculus
Week 4:	Numerical Differentiation and Integration
Week 5:	Solving Nonlinear Equations
Week 6:	Probability Theory Basics
Week 7:	Random Variables and Distributions
Week 8:	Statistics for Data Science
Week 9:	Eigenvalues and Eigenvectors
Week 10:	Simulation and Monte Carlo Method
Week 11:	Data Fitting and Regression
Week 12:	Optimization Techniques
Week 13:	Machine Learning Fundamentals



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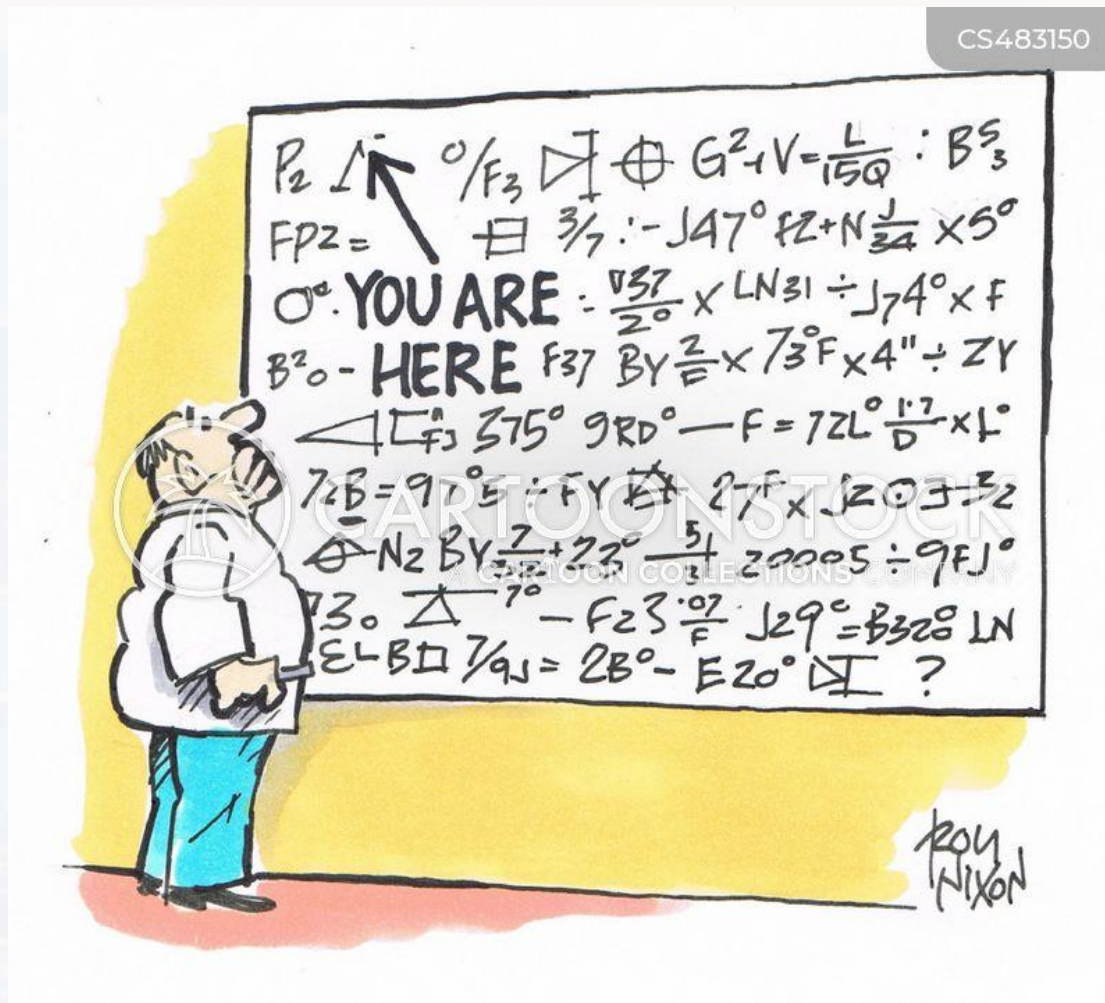


Outline

- Recap: Calculus
- Gradient
- Line Integrals
- Divergence
- Curl



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Outline

- Recap: Calculus

- Gradient

- Line Integrals

- Divergence

- Curl



derivatives

motivation:

- optimization algorithms (gradient descent and related)
- ANNs learn via **backpropagation** → chain rule
- approximation methods (Taylor series)
- maximum entropy distributions (data analysis, data modelling)
- error estimation and error propagation
- and more...



derivatives

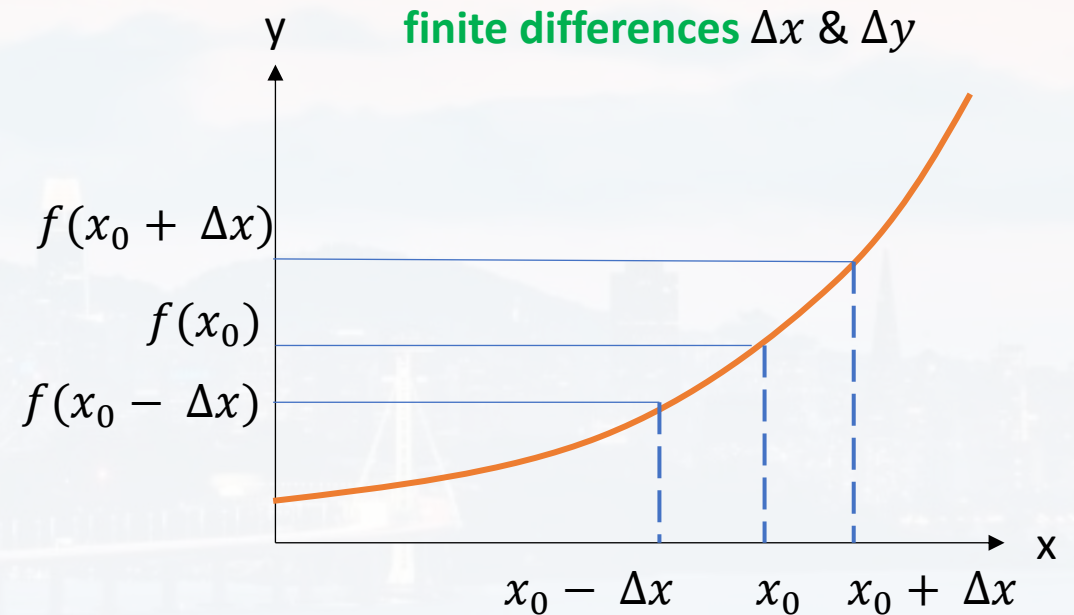
slope of a function at $x = x_0$

forward derivative

$$\left. \frac{df^+}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

backward derivative

$$\left. \frac{df^-}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0) - f(x_0 - \Delta x)}{\Delta x}$$



average converges faster

$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$



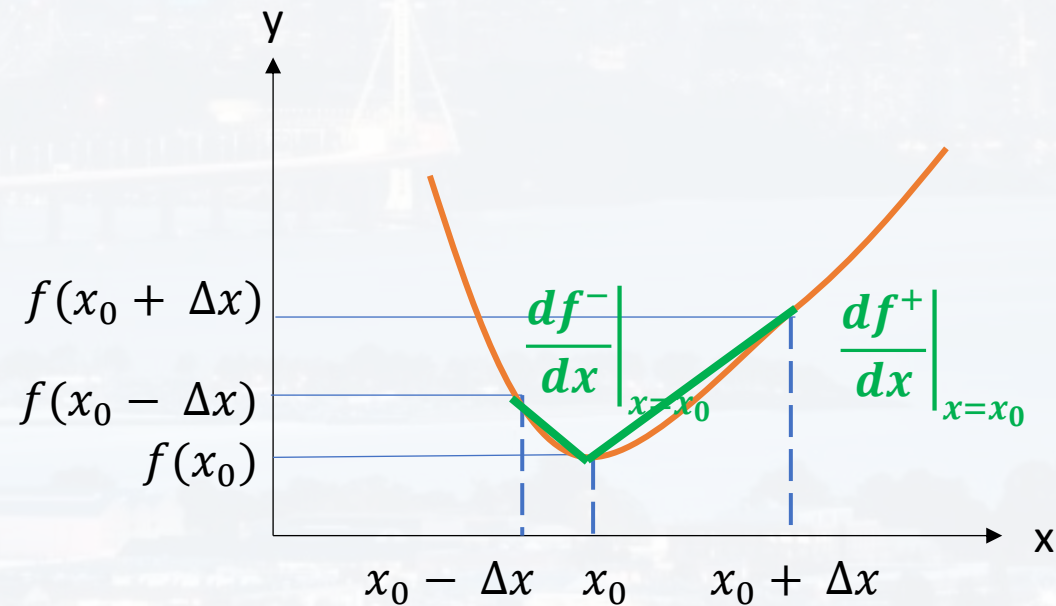
derivatives

$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

change of the slope of a function at $x = x_0$, aka curvature

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} - \left. \frac{df^-}{dx} \right|_{x=x_0} \right)$$





derivatives

$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

change of the slope of a function at $x = x_0$, aka *curvature*

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} - \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - 2f(x_0) + f(x_0 - \Delta x)}{\Delta x^2}$$

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - 2f(x_0) + f(x_0 - \Delta x)}{\Delta x^2}$$

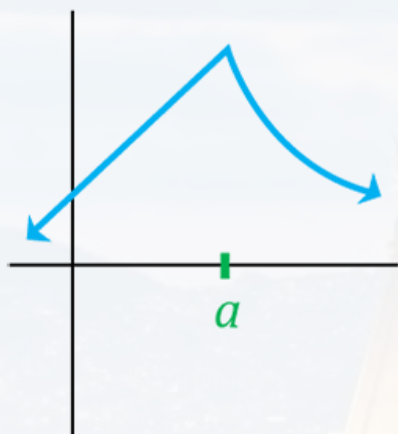
2nd derivative at $x = x_0$

...and so on

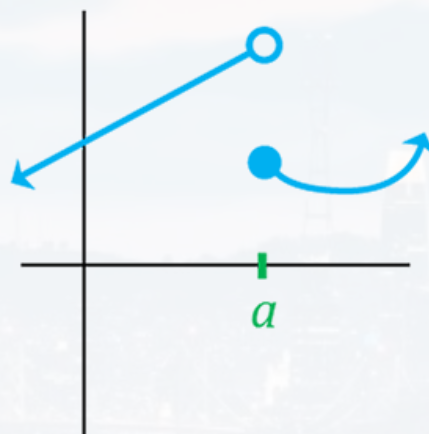


derivatives

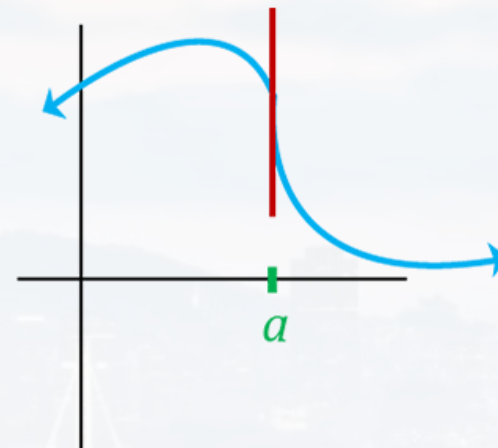
derivatives are not always defined:



Cusp / Corner

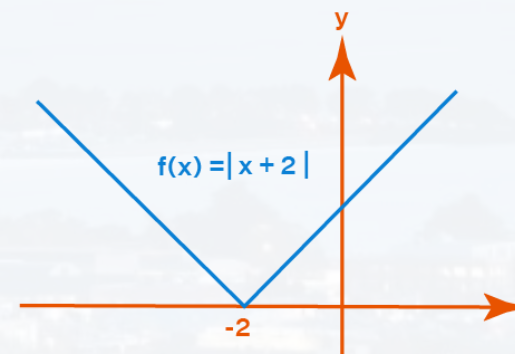


Discontinuous



Vertical Tangent

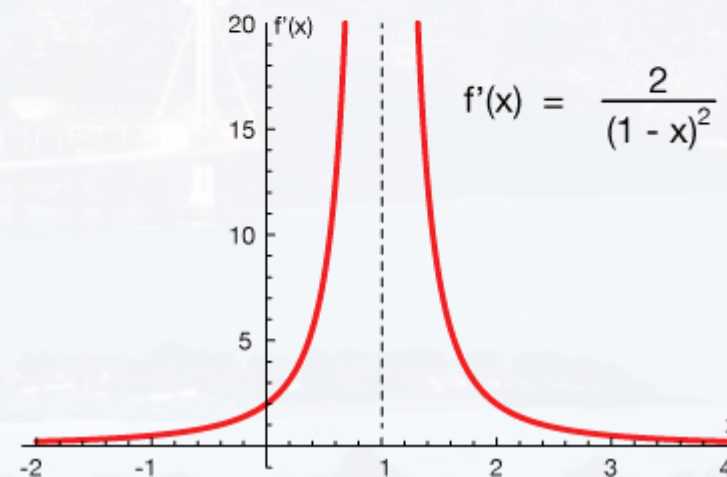
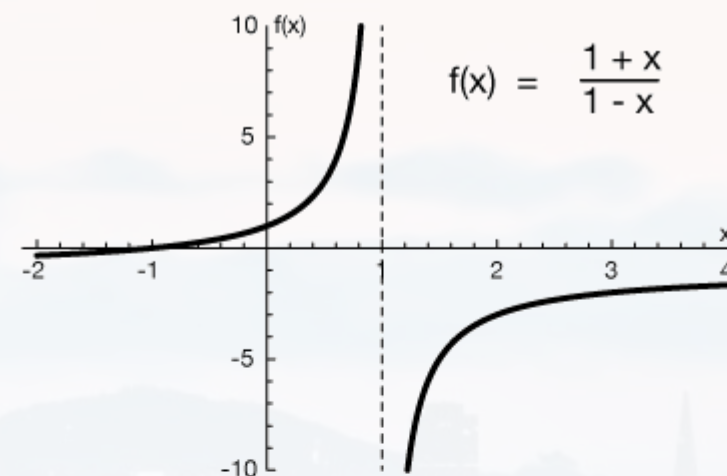
function needs to be **continuous and differentiable**





derivatives

derivatives are not always defined:



function needs to be **continuous and differentiable**



derivatives

example: $f(x) = \sqrt{x}$

$$\begin{aligned}\left. \frac{df}{dx} \right|_{x=x_0} &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{x_0 + \Delta x} - \sqrt{x_0 - \Delta x}}{2\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{(\sqrt{x_0 + \Delta x} - \sqrt{x_0 - \Delta x})(\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})}{2\Delta x (\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})} \\ &= \lim_{\Delta x \rightarrow 0} \frac{x_0 + \Delta x - x_0 + \Delta x}{2\Delta x (\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2\Delta x}{2\Delta x (\sqrt{x_0 + \Delta x} + \sqrt{x_0 - \Delta x})} = \frac{1}{2\sqrt{x_0}}\end{aligned}$$



derivatives

$$a \in \mathbb{C}$$

$$n \in \mathbb{R}$$

rules: $\frac{d}{dx} ax^n = a n x^{n-1}$

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

sum rule: derivatives are linear

$$\frac{d}{dx} [f(x)g(x)] = g(x) \frac{d}{dx} f(x) + f(x) \frac{d}{dx} g(x)$$

product rule

$$\frac{d}{dx} \textcolor{teal}{f}[\textcolor{blue}{g}(x)] = \frac{df(x)}{dg(x)} \frac{d}{dx} g(x)$$

chain rule

outer derivative **inner derivative**



derivatives

$$a \in \mathbb{C}$$

$$n \in \mathbb{R}$$

special derivatives

$$\frac{d}{dx} e^x = e^x$$

the actual definition of e

$$\frac{d}{dx} b^x = \ln(b) b^x$$

$$b > 0$$

$$\frac{d}{dx} \log_b(x) = \frac{1}{x \ln(b)}$$

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$



derivatives

$$a \in \mathbb{C}$$

$$n \in \mathbb{R}$$

$$\frac{d}{dx} ax^n = a n x^{n-1}$$

$$\frac{d}{dx} 3x^5 = 15x^4$$

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

$$\frac{d}{dx} [3x^5 - 2x] = 15x^4 - 2$$

$$\frac{d}{dx} [f(x)g(x)] = g(x) \frac{d}{dx} f(x) + f(x) \frac{d}{dx} g(x)$$

$$\frac{d}{dx} [3x^5 \sin(x)] = 15x^4 \sin(x) + 3x^5 \cos(x)$$

$$\frac{d}{dx} f[g(x)] = \frac{df(x)}{dg(x)} \frac{d}{dx} g(x)$$

outer derivative inner derivative

$$\frac{d}{dx} \sin(3x^5) = \cos(3x^5) 15x^4$$

outer derivative inner derivative



derivatives

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

$$n = 0: \quad f(x) \approx f(x_0)$$

$$n = 1: \quad f(x) \approx f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0) \quad \text{tangent on } f \text{ at } x = x_0$$

$$\frac{f(x) - f(x_0)}{x - x_0} \approx \left. \frac{df}{dx} \right|_{x=x_0}$$

$$\left. \frac{df}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

definition of the 1st derivative!



derivatives

Taylor Series

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$$n = 2: \quad f(x) \approx f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0) + \frac{1}{2} \left. \frac{d^2 f}{dx^2} \right|_{x=x_0} (x - x_0)^2$$



derivatives

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

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exercise:

- write down the Taylor Series of $\sin(x)$, $\cos(x)$ and e^x at $x_0 = 0$
- express all three series as an infinite sum
- try to combine all three equations by introducing a new mathematical object i which only property is $i^2 = -1$

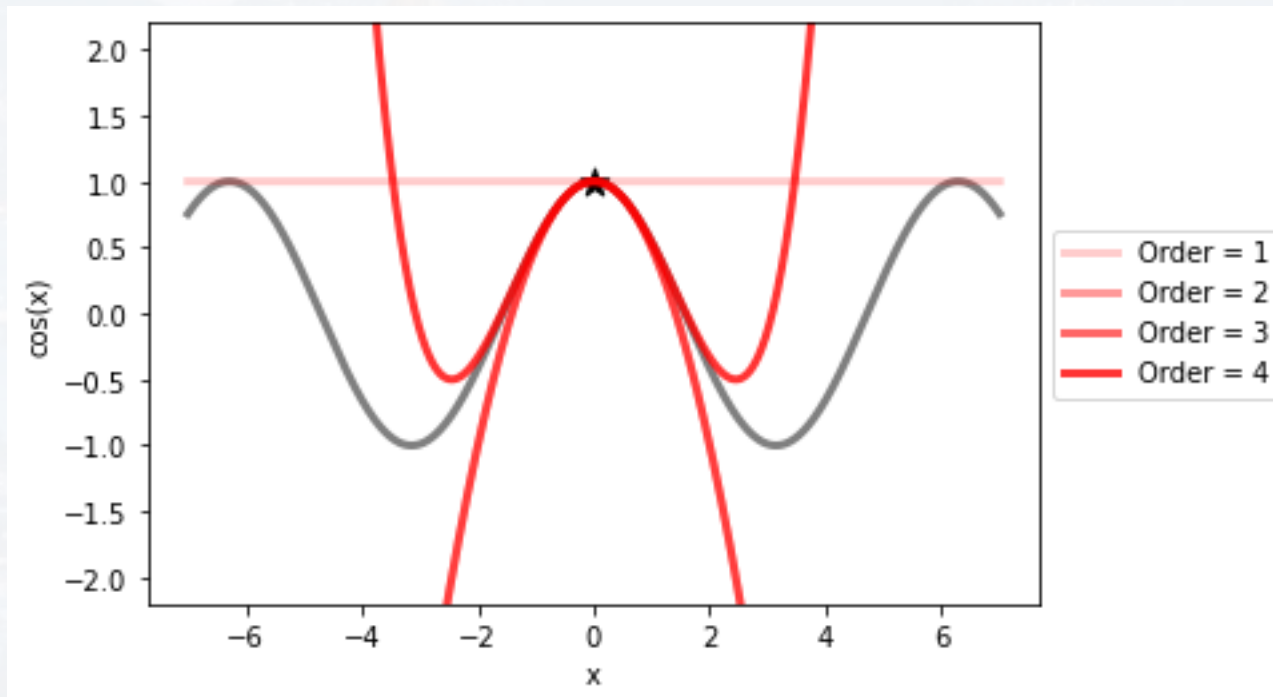


derivatives

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

run the function `PlotTaylorSeries.py`



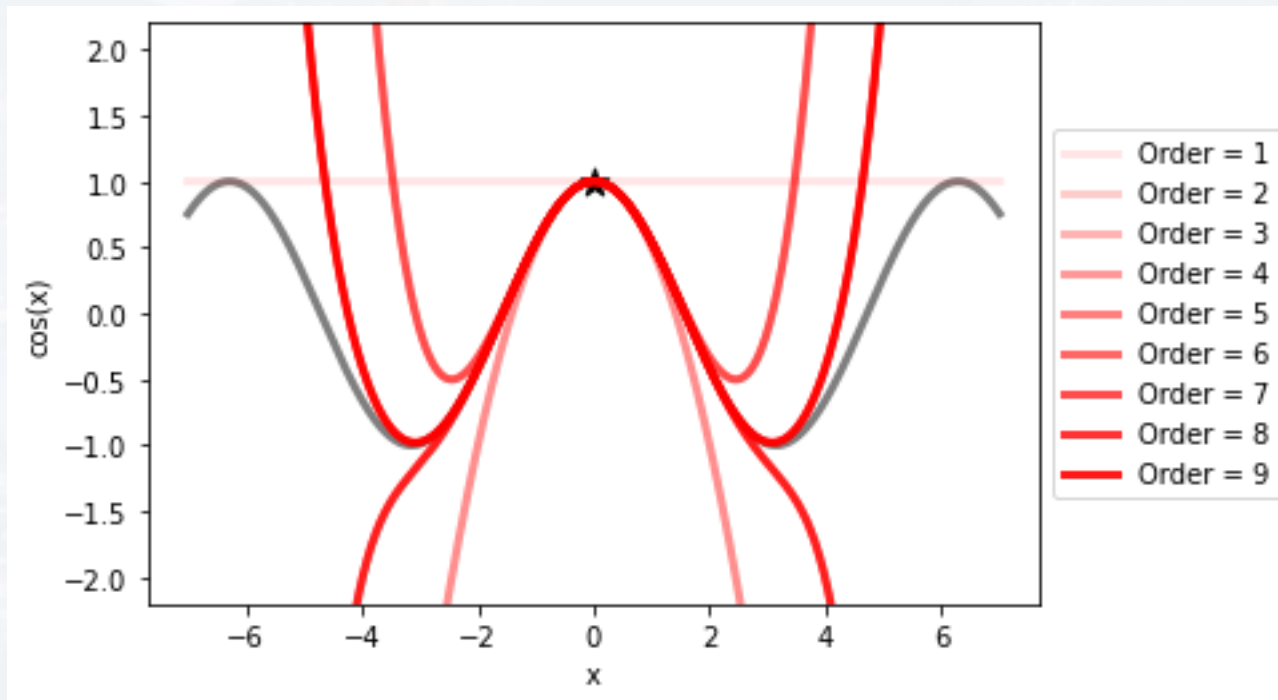


derivatives

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

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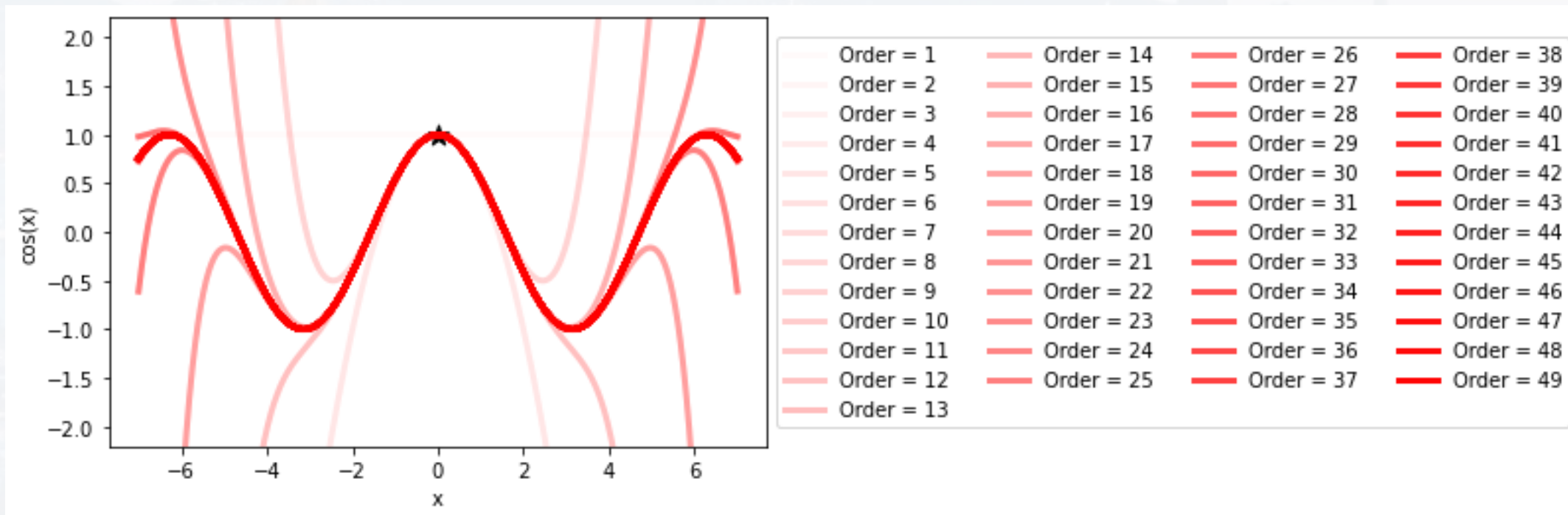


derivatives

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x=x_0} (x - x_0)^n$$

run the function `PlotTaylorSeries.py`





integrals

motivation:

- deriving probabilities from likelihood functions
- normalization tools
- calculating volumes, areas, flow, energy, etc....
- sums $\rightarrow$ integral



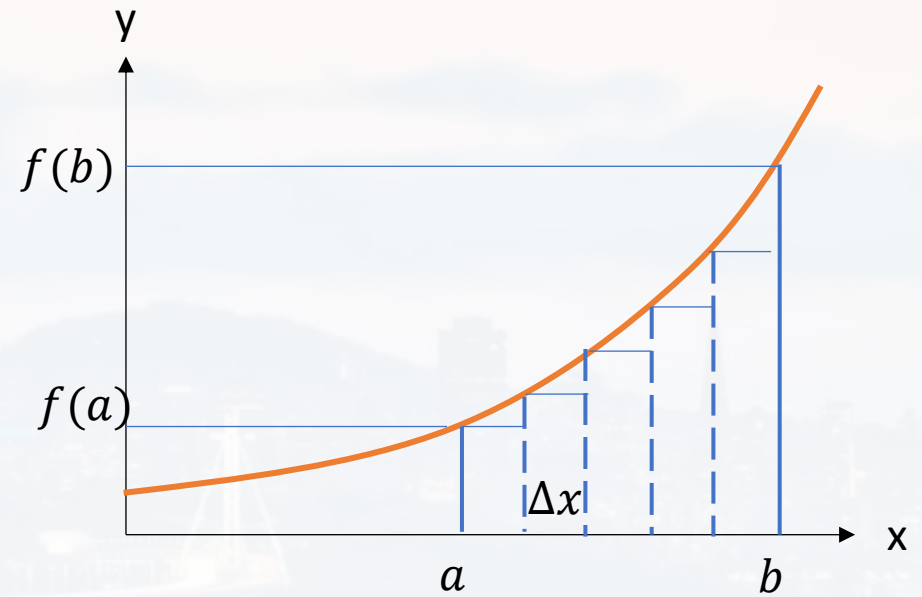
integrals

area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx$$





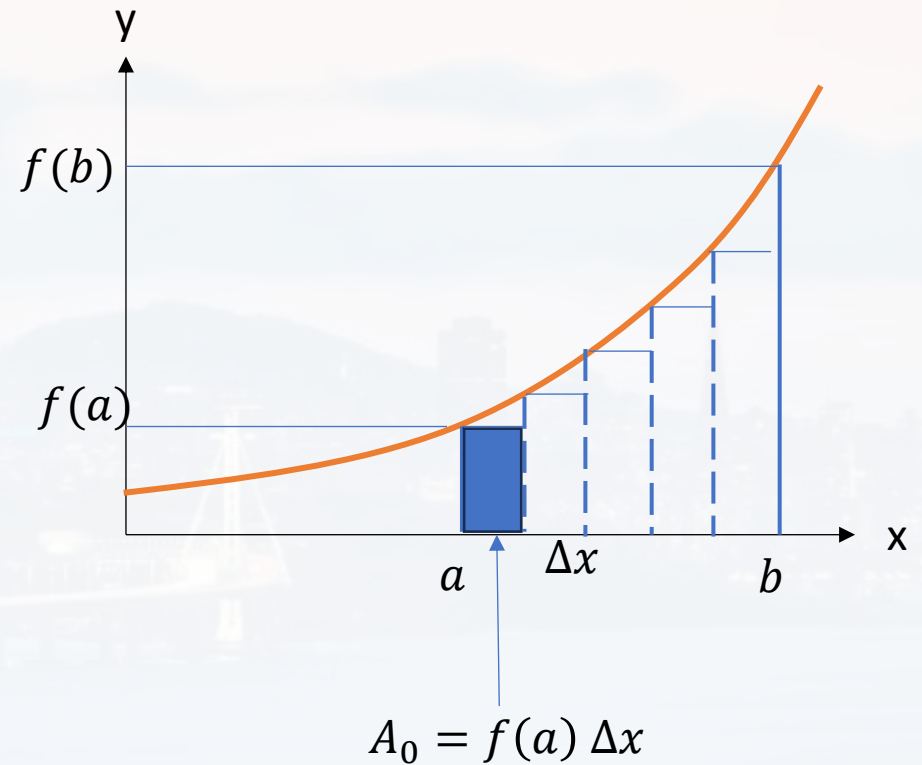
integrals

area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x$$





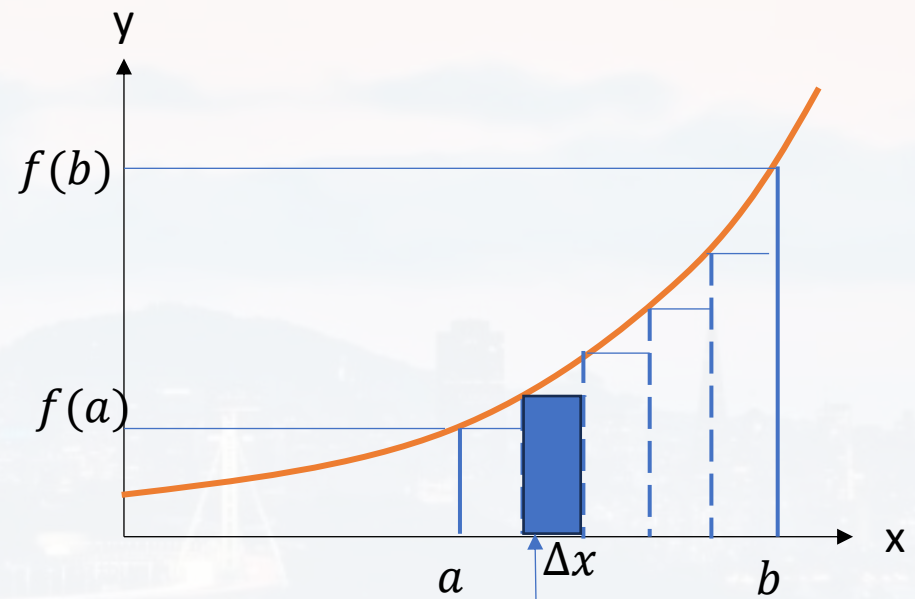
integrals

area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x + f(a + \Delta x) \Delta x$$



$$A_1 = f(a + \Delta x) \Delta x$$



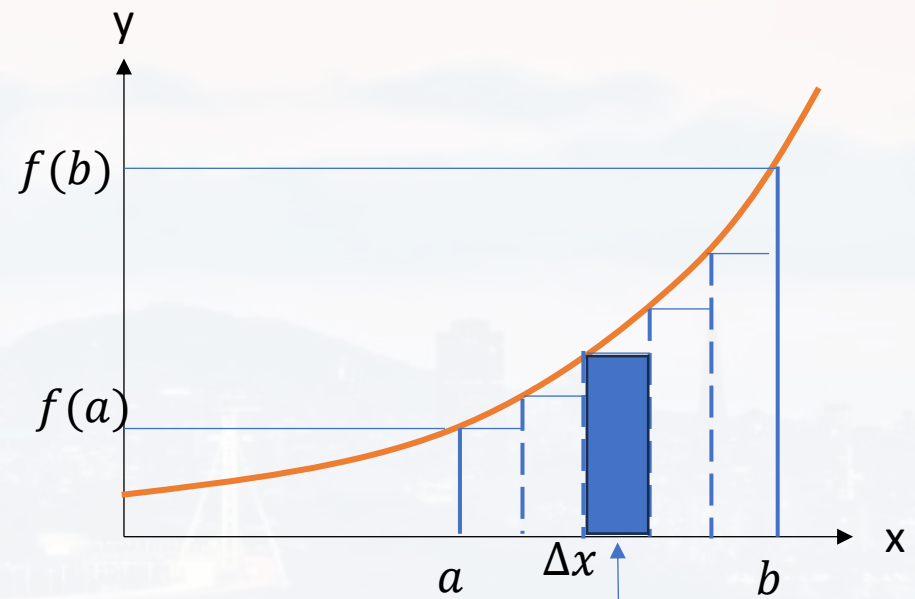
integrals

area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x + f(a + \Delta x) \Delta x + f(a + 2\Delta x) \Delta x$$



$$A_2 = f(a + 2\Delta x) \Delta x$$



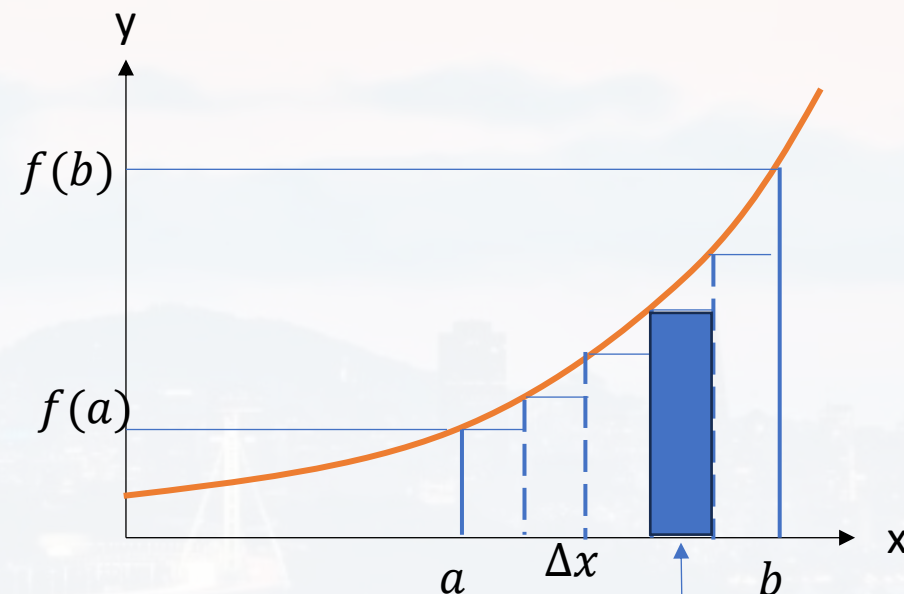
integrals

area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a) \Delta x + f(a + \Delta x) \Delta x + f(a + 2\Delta x) \Delta x + f(a + 3\Delta x) \Delta x$$



$$A_3 = f(a + 3\Delta x) \Delta x$$



integrals

area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$A_{tot} \approx f(a + \mathbf{0}\Delta x) \Delta x$$

$$+ f(a + \mathbf{1}\Delta x) \Delta x$$

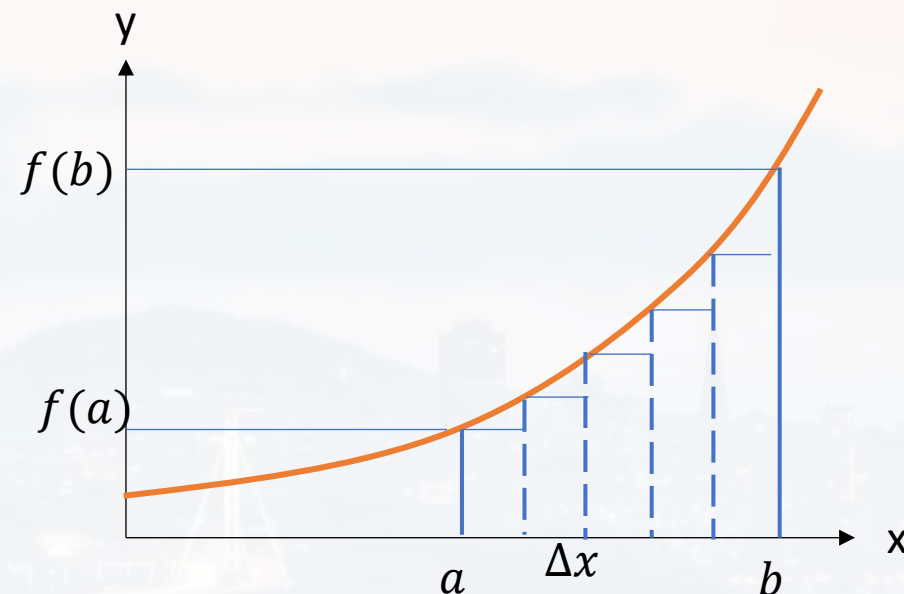
$$+ f(a + \mathbf{2}\Delta x) \Delta x$$

$$+ f(a + \mathbf{3}\Delta x) \Delta x$$

$$+ f(a + \mathbf{4}\Delta x) \Delta x$$

$$+ \dots$$

$$= \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$





integrals

area under a curve (1D)

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

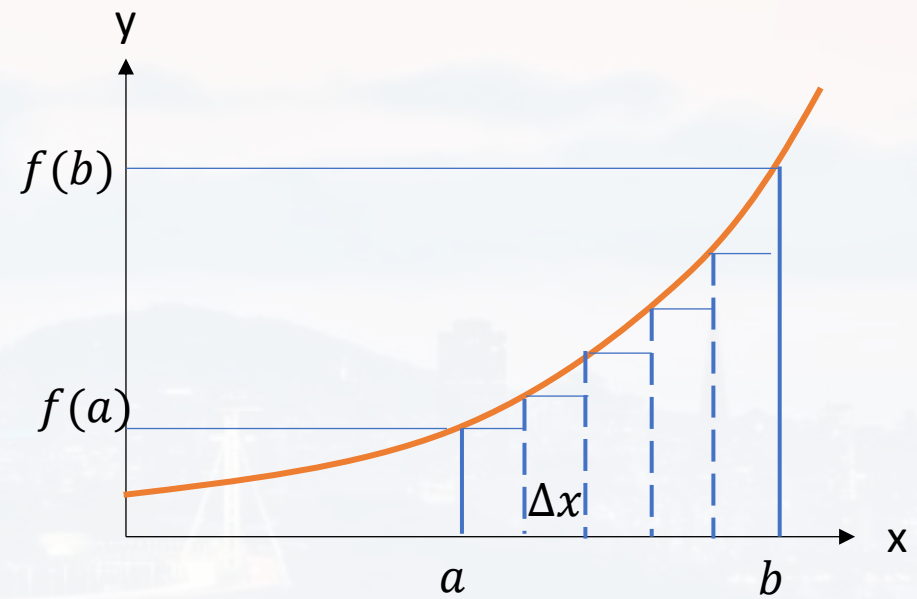
$$N = \frac{b - a}{\Delta x}$$

more accurate:

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i + 1) \Delta x)] \frac{\Delta x}{2}$$

error (for large N):

$$\varepsilon = -\frac{(b - a)^2}{12 N^2} [f'(b) - f'(a)] + O(N^{-3})$$



trapezoidal rule



integrals

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i+1) \Delta x)] \frac{\Delta x}{2} \quad N = \frac{b-a}{\Delta x}$$

example: $f(x) = x^2$

$$\int_a^b x^2 dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [(a + i \Delta x)^2 + (a + (i+1) \Delta x)^2] \frac{\Delta x}{2}$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [2a^2 + i^2 \Delta x^2 + 2ai \Delta x + a^2 + (i+1)^2 \Delta x^2 + 2a(i+1) \Delta x] \frac{\Delta x}{2}$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [2a^2 + i^2 \Delta x^2 + 2ai \Delta x + i^2 \Delta x^2 + \Delta x^2 + 2i \Delta x^2 + 2ai \Delta x + 2a \Delta x] \frac{\Delta x}{2}$$



integrals

example: $f(x) = x^2$

$$N = \frac{b - a}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [2a^2 + i^2 \Delta x^2 + 2ai\Delta x + i^2 \Delta x^2 + \Delta x^2 + 2i\Delta x^2 + 2ai\Delta x + 2a\Delta x] \frac{\Delta x}{2}$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \sum_{i=0}^{N-1} i^2 + 2a\Delta x^2 \sum_{i=0}^{N-1} i + \frac{\Delta x^3}{2} N + \Delta x^3 \sum_{i=0}^{N-1} i + a\Delta x^2 N \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \sum_{i=0}^{N-1} i^2 + 2a\Delta x^2 \sum_{i=0}^{N-1} i + \Delta x^3 \sum_{i=0}^{N-1} i \right]$$



integrals

example: $f(x) = x^2$

$$N = \frac{b - a}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \sum_{i=0}^{N-1} i^2 + 2a\Delta x^2 \sum_{i=0}^{N-1} i + \Delta x^3 \sum_{i=0}^{N-1} i \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{N(N+1)(2N+1)}{6} + a\Delta x^2 N(N+1) + \Delta x^3 \frac{N(N+1)}{2} \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{N(N+1)(2N+1)}{6} + a\Delta x^2 N(N+1) \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{(N^2 + N)(2N+1)}{6} + a\Delta x^2 N^2 + a\Delta x^2 N \right]$$



integrals

example: $f(x) = x^2$

$$N = \frac{b - a}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{(N^2 + N)(2N + 1)}{6} + a \Delta x^2 N^2 + a \Delta x^2 N \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 N + \Delta x^3 \frac{2N^3 + 3N^2 + N}{6} + a \Delta x^2 N^2 \right]$$

$$= \lim_{\Delta x \rightarrow 0} \left[\Delta x a^2 \frac{b - a}{\Delta x} + \Delta x^3 \frac{2 \left(\frac{b - a}{\Delta x} \right)^3 + 3 \left(\frac{b - a}{\Delta x} \right)^2 + \frac{b - a}{\Delta x}}{6} + a \Delta x^2 \left(\frac{b - a}{\Delta x} \right)^2 \right]$$

$$= a^2(b - a) + \frac{(b - a)^3}{3} + a(b - a)^2 = \frac{(b - a)^3}{3}$$

$$\int_a^b x^2 dx = \frac{(b - a)^3}{3}$$



integrals

$$a, c \in \mathbb{C}$$
$$n \in \mathbb{R}$$

rules: $\int \frac{d}{dx} f(x) dx = \int df(x) = f(x) + c$

therefore: an integral is
an **anti derivative!**

$$\int ax^n dx = a \frac{1}{n+1} x^{n+1} + c \quad n \neq -1$$

$$\int f(x) + g(x) dx = \int f(x) dx + \int g(x) dx + c$$

sum rule: integrals are linear

$$\int f(x) \frac{d}{dx} g(x) dx = f(x)g(x) + \int \frac{d}{dx} f(x) \cdot g(x) dx + c$$

product rule



integrals

special integrals

$$\int e^x dx = e^x + c$$

$$\int a^x dx = \frac{a^x}{\ln(a)} + c$$

$$\int \frac{1}{x} dx = \ln(|x|) + c$$

$$\int \log_b(x) dx = x \log_b(x) - \frac{x}{\ln(b)} + c$$

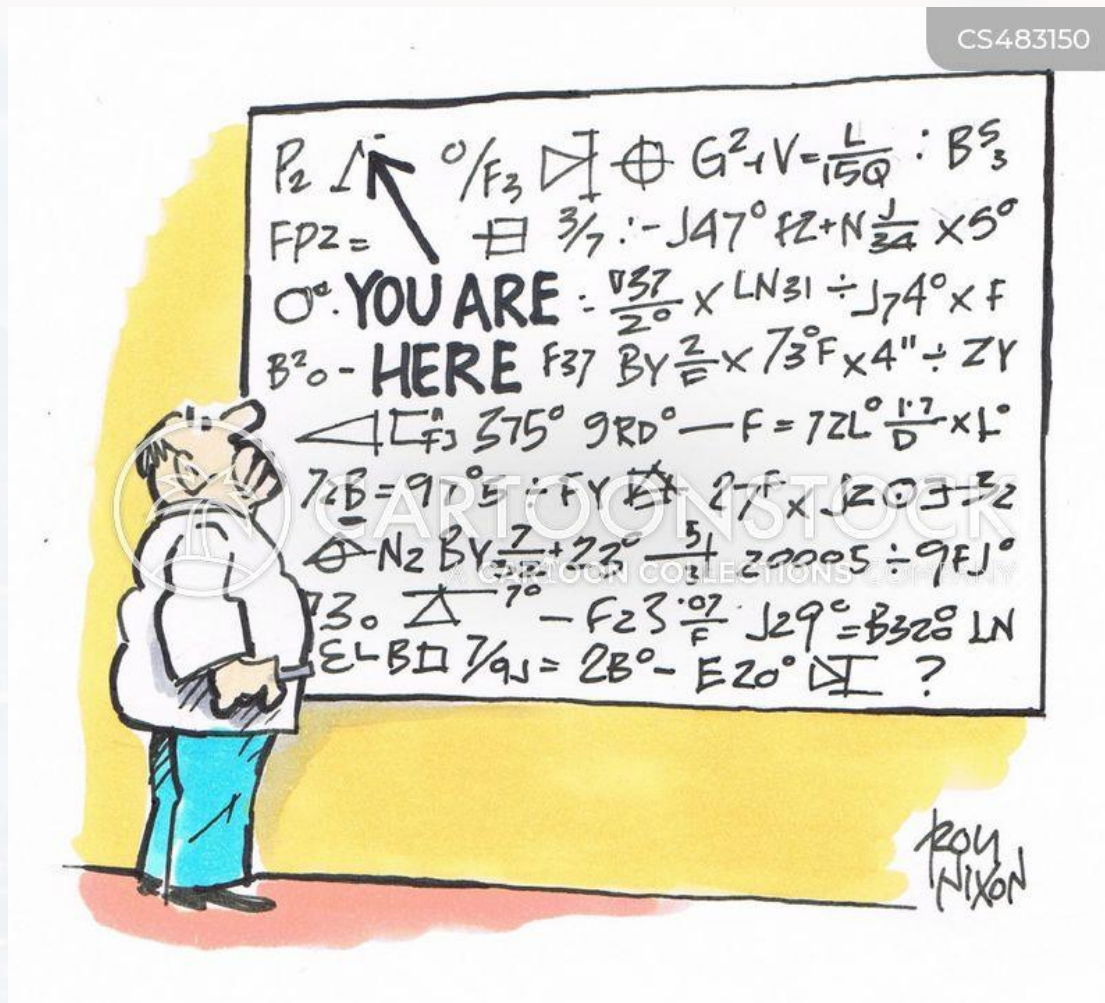
$$\int \cos(x) dx = \sin(x) + c$$

$$\int \sin(x) dx = -\cos(x) + c$$

$$a, c \in \mathbb{C}$$
$$n \in \mathbb{R}$$



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Outline

- Recap: Calculus
- **Gradient**
- Line Integrals
- Divergence
- Curl

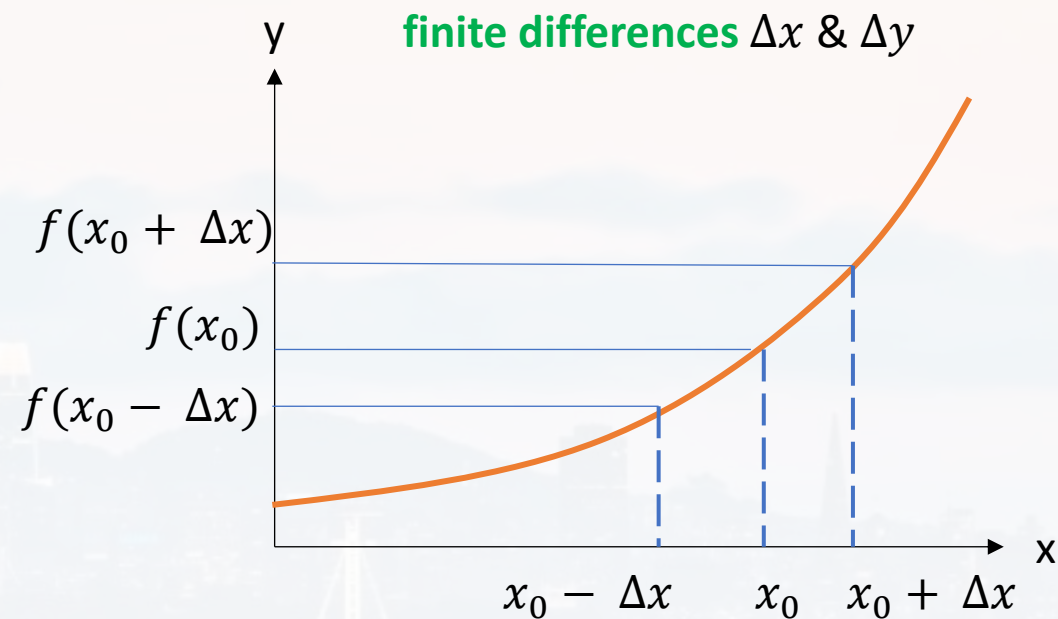
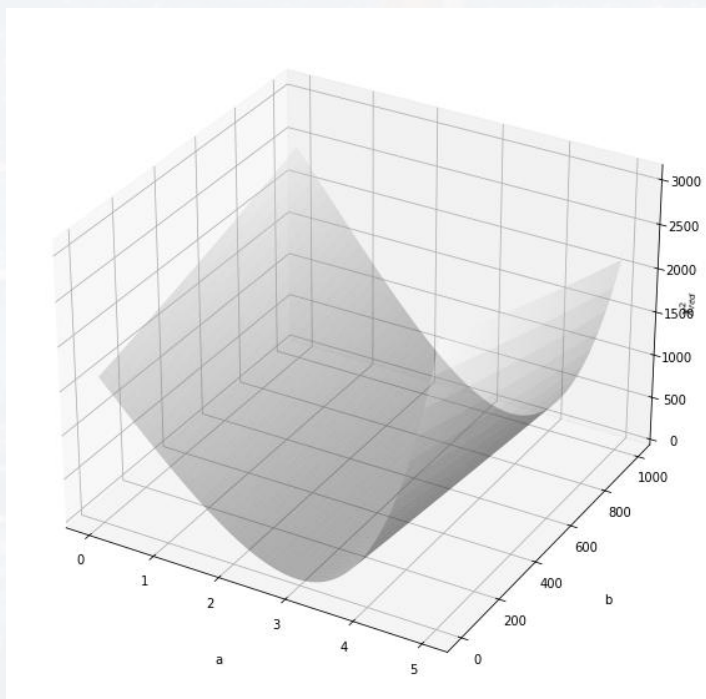


$$\left. \frac{df}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

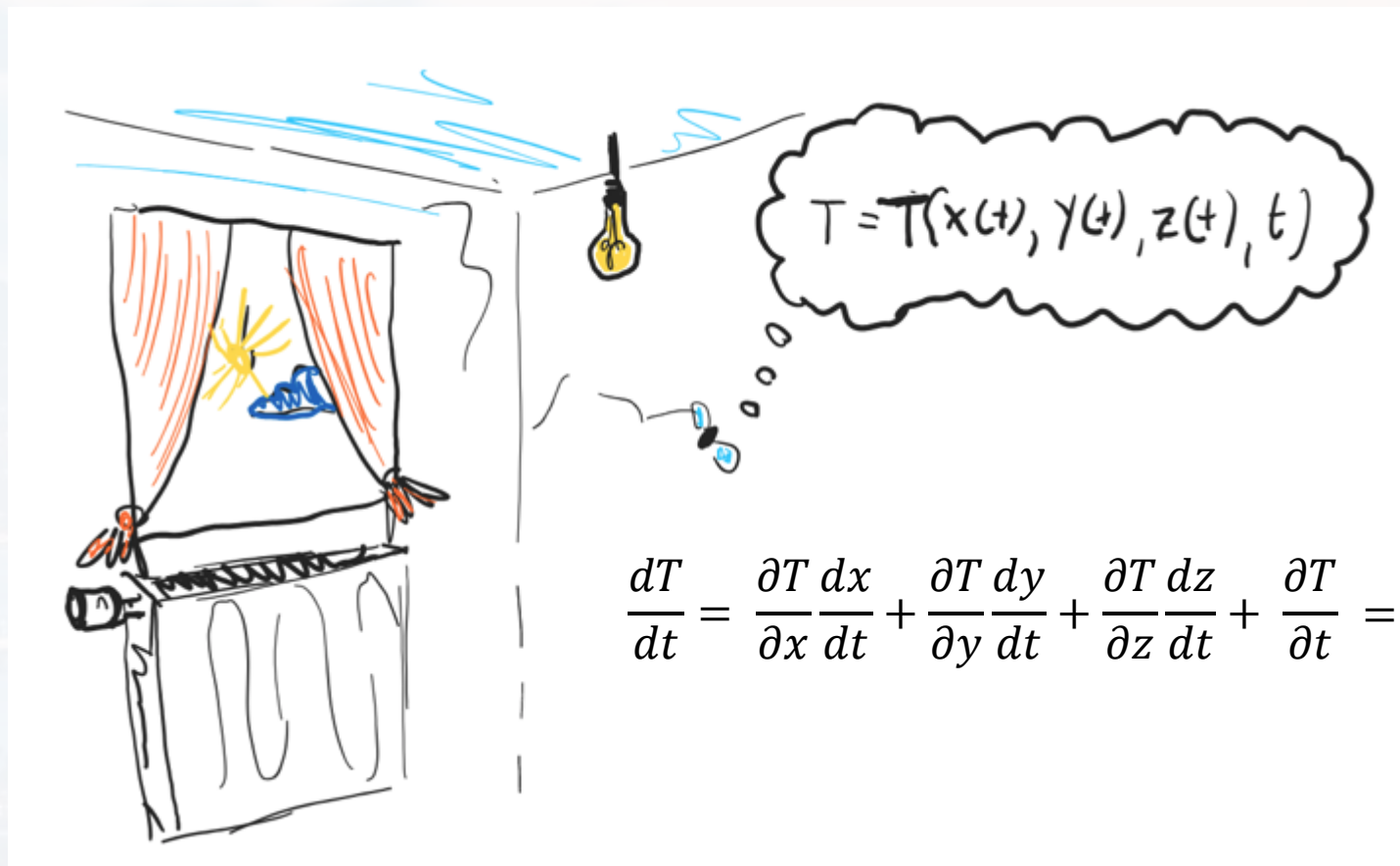
1st derivative of a 1D function

but the world is **multidimensional**





temperature profile in space and time



T : temperature

$$\frac{dT}{dt} = \frac{\partial T}{\partial x} \frac{dx}{dt} + \frac{\partial T}{\partial y} \frac{dy}{dt} + \frac{\partial T}{\partial z} \frac{dz}{dt} + \frac{\partial T}{\partial t} = \begin{pmatrix} \frac{\partial T}{\partial x} \\ \frac{\partial T}{\partial y} \\ \frac{\partial T}{\partial z} \end{pmatrix} \circ \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{pmatrix} + \frac{\partial T}{\partial t}$$

$$= \text{grad}(T) \circ \vec{v}$$

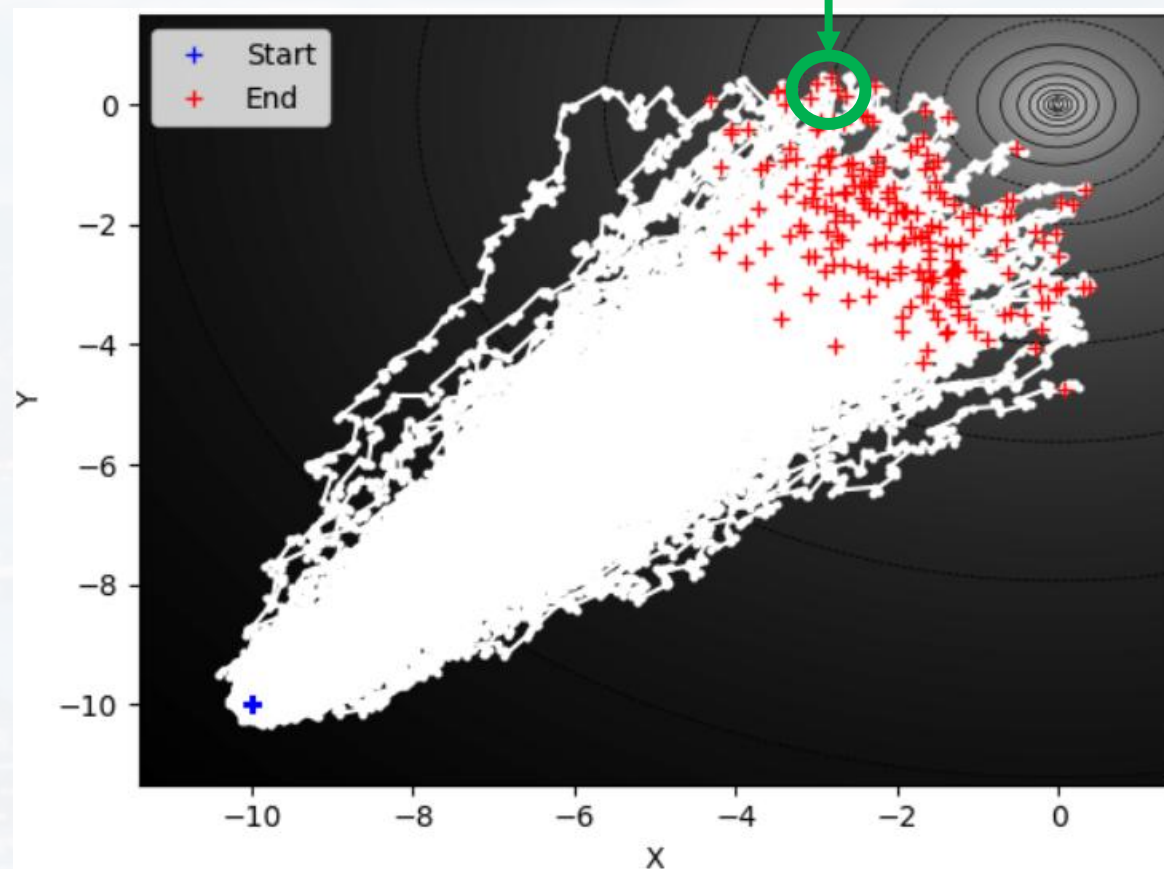
If T doesn't change
with time!



concentration profile in space and time

E. Coli

c : concentration



$$\frac{dc}{dt} = \frac{\partial c}{\partial x} \frac{dx}{dt} + \frac{\partial c}{\partial y} \frac{dy}{dt} + \frac{\partial c}{\partial z} \frac{dz}{dt} + \frac{\partial c}{\partial t}$$

$$= \begin{pmatrix} \frac{\partial c}{\partial x} \\ \frac{\partial c}{\partial y} \\ \frac{\partial c}{\partial z} \end{pmatrix} \circ \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{pmatrix} + \frac{\partial c}{\partial t}$$

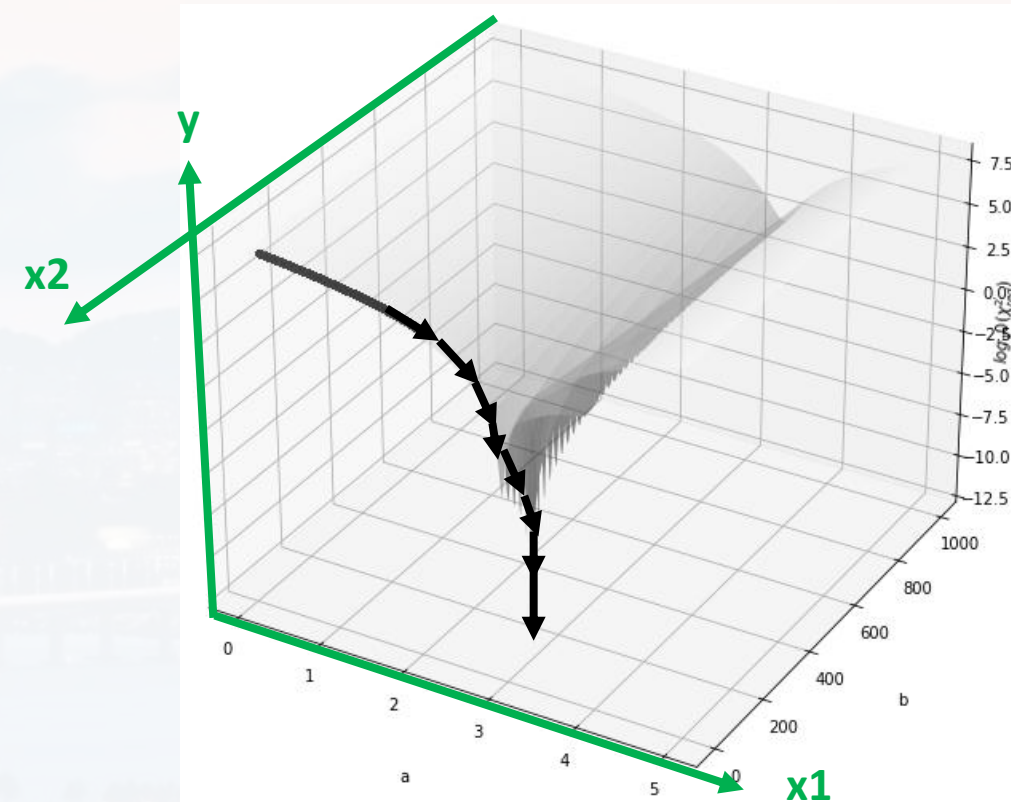
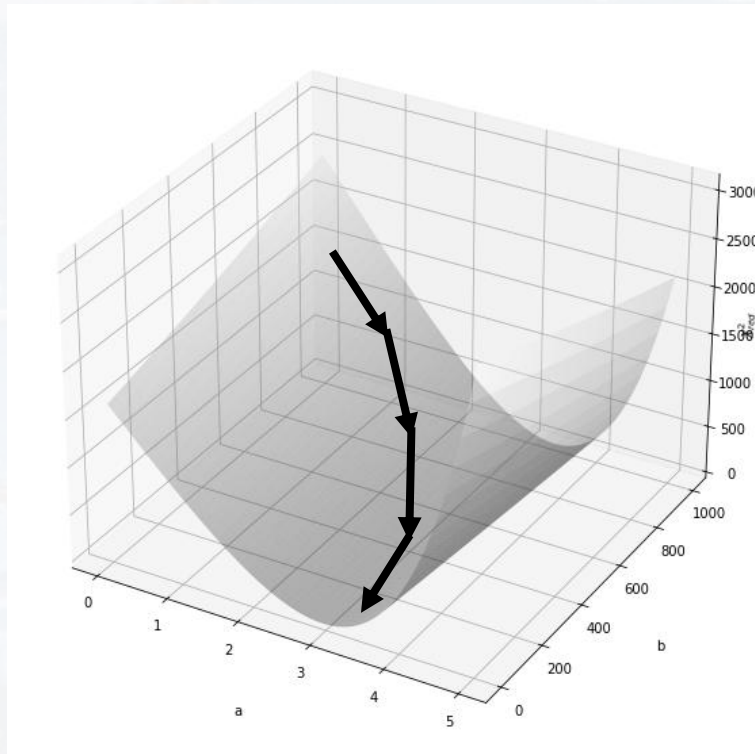
$$= \text{grad}(c) \circ \vec{v}$$

If c doesn't change with time!



$$\left. \frac{\partial y}{\partial x_1} \right|_{x_1^*; x_2^*} \approx \frac{y(x_1^* + \Delta x_1, x_2^*) - y(x_1^* - \Delta x_1, x_2^*)}{2\Delta x_1}$$

$$\left. \frac{\partial y}{\partial x_2} \right|_{x_1^*; x_2^*} \approx \frac{y(x_1^*, x_2^* + \Delta x_2) - y(x_1^*, x_2^* - \Delta x_2)}{2\Delta x_2}$$





$$\left. \frac{\partial y}{\partial x_1} \right|_{x_1^*; x_2^*; \dots; x_N^*} \approx \frac{y(x_1^* + \Delta x_1, x_2^*, \dots, x_N^*) - y(x_1^* - \Delta x_1, x_2^*, \dots, x_N^*)}{2\Delta x_1}$$

$$\left. \frac{\partial y}{\partial x_2} \right|_{x_1^*; x_2^*; \dots; x_N^*} \approx \frac{y(x_1^*, x_2^* + \Delta x_2, \dots, x_N^*) - y(x_1^*, x_2^* - \Delta x_2, \dots, x_N^*)}{2\Delta x_2}$$

⋮

$$\left. \frac{\partial y}{\partial x_i} \right|_{x_1^*; x_2^*; \dots; x_N^*} \approx \frac{y(\dots, x_i^* + \Delta x_i, \dots, x_N^*) - y(\dots, x_i^* - \Delta x_i, \dots, x_N^*)}{2\Delta x_i}$$

⋮

$$\left. \frac{\partial y}{\partial x_N} \right|_{x_1^*; x_2^*; \dots; x_N^*} \approx \frac{y(x_1^*, x_2^*, \dots, x_N^* + \Delta x_N) - y(x_1^*, x_2^*, \dots, x_N^* - \Delta x_N)}{2\Delta x_N}$$

$$\begin{pmatrix} \left. \frac{\partial y}{\partial x_1} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \vdots \\ \left. \frac{\partial y}{\partial x_i} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \vdots \\ \left. \frac{\partial y}{\partial x_N} \right|_{x_1^*; x_2^*; \dots; x_N^*} \end{pmatrix}$$

$= \text{grad}(y)_x$
gradient of y wrt x



$$\begin{pmatrix} \left. \frac{\partial y}{\partial x_1} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \dots \\ \left. \frac{\partial y}{\partial x_i} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \dots \\ \left. \frac{\partial y}{\partial x_N} \right|_{x_1^*; x_2^*; \dots; x_N^*} \end{pmatrix} = \text{grad}(y)_x \equiv \vec{\nabla} y$$

The gradient of a function f is a **vector**,
because the derivatives have a direction!

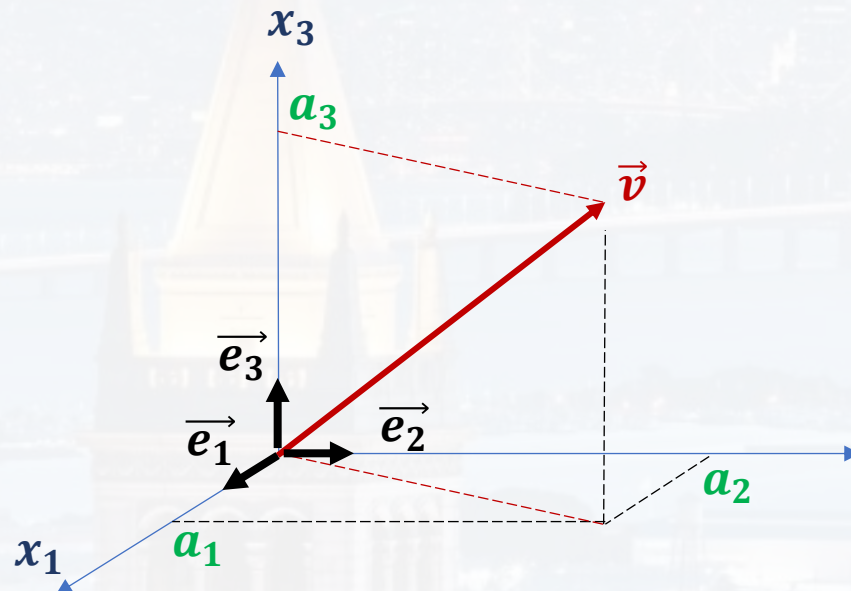
unit vectors $\vec{e}_1$, $\vec{e}_2$ and $\vec{e}_3$

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

length 1 (normalized)
mutually orthogonal } ortho normal

$$\vec{v} = a_1 \vec{e}_1 + a_2 \vec{e}_2 + a_3 \vec{e}_3$$

$$\vec{v} = a_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + a_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$





$$\begin{pmatrix} \left. \frac{\partial y}{\partial x_1} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \dots \\ \left. \frac{\partial y}{\partial x_i} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \dots \\ \left. \frac{\partial y}{\partial x_N} \right|_{x_1^*; x_2^*; \dots; x_N^*} \end{pmatrix} = \text{grad}(y)_x \equiv \vec{\nabla} y$$

The gradient of a function f is **a vector**,
because the derivatives have a direction!

unit vectors $\vec{e}_1$, $\vec{e}_2$ and $\vec{e}_3$

$$\vec{v} = \mathbf{a}_1 \vec{e}_1 + \mathbf{a}_2 \vec{e}_2 + \mathbf{a}_3 \vec{e}_3$$

$$\vec{\nabla} y = \frac{\partial y}{\partial x_1} \vec{e}_1 + \frac{\partial y}{\partial x_2} \vec{e}_2 + \dots \frac{\partial y}{\partial x_i} \vec{e}_i \dots + \frac{\partial y}{\partial x_N} \vec{e}_N = \left(\frac{\partial}{\partial x_1} \vec{e}_1 + \frac{\partial}{\partial x_2} \vec{e}_2 + \dots \frac{\partial}{\partial x_i} \vec{e}_i \dots + \frac{\partial}{\partial x_N} \vec{e}_N \right) y$$



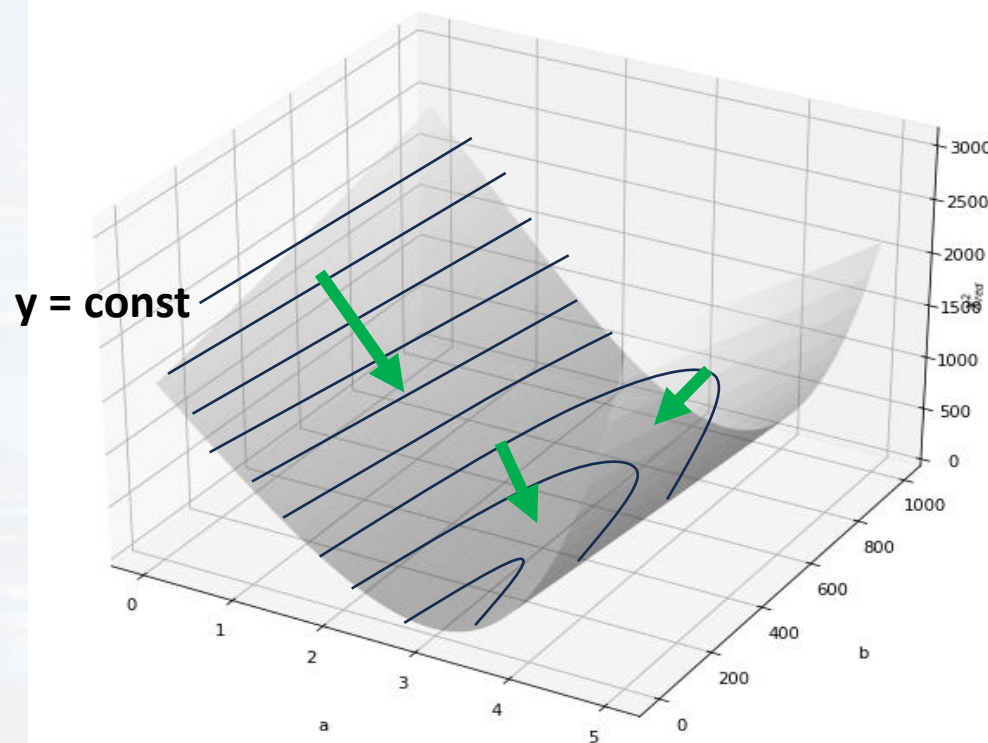
$$\begin{pmatrix} \left. \frac{\partial y}{\partial x_1} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \dots \\ \left. \frac{\partial y}{\partial x_i} \right|_{x_1^*; x_2^*; \dots; x_N^*} \\ \dots \\ \left. \frac{\partial y}{\partial x_N} \right|_{x_1^*; x_2^*; \dots; x_N^*} \end{pmatrix} = \text{grad}(y)_x \equiv \vec{\nabla} y$$

$$\vec{\nabla} y = \frac{\partial y}{\partial x_1} \vec{e}_1 + \frac{\partial y}{\partial x_2} \vec{e}_2 + \dots \frac{\partial y}{\partial x_i} \vec{e}_i \dots + \frac{\partial y}{\partial x_N} \vec{e}_N$$

$$= \left(\frac{\partial}{\partial x_1} \vec{e}_1 + \frac{\partial}{\partial x_2} \vec{e}_2 + \dots \frac{\partial}{\partial x_i} \vec{e}_i \dots + \frac{\partial}{\partial x_N} \vec{e}_N \right) y$$

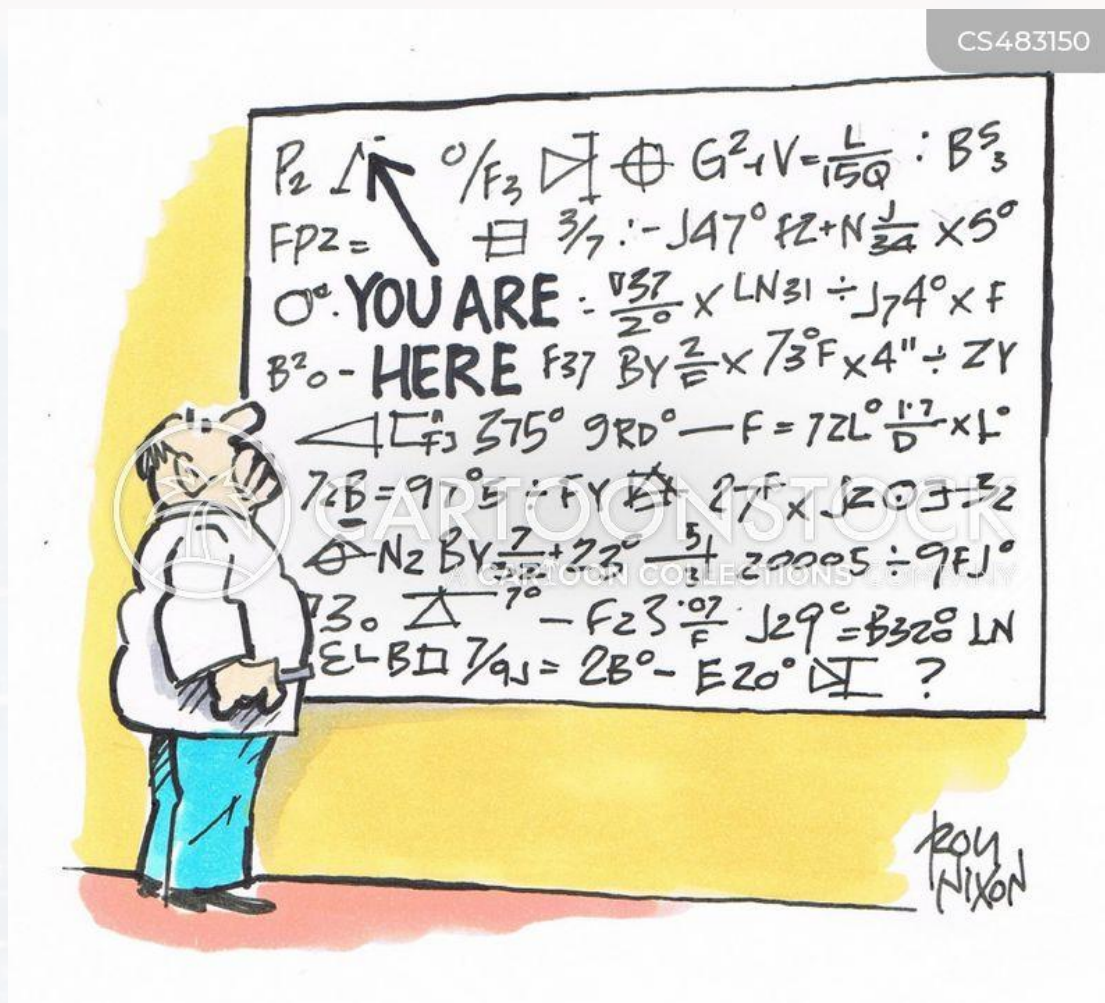
gradient: vector containing all the partial derivatives at point $P = P(x_1^*; x_2^*; \dots; x_N^*)$

grad(y)_x is perpendicular to $y = \text{const}$





CS483150



Outline

- Recap: Calculus
- Gradient
- **Line Integrals**
- Divergence
- Curl



We will be integrating over a path:
line integral

If the path is **closed**
(start and endpoint
are identical):

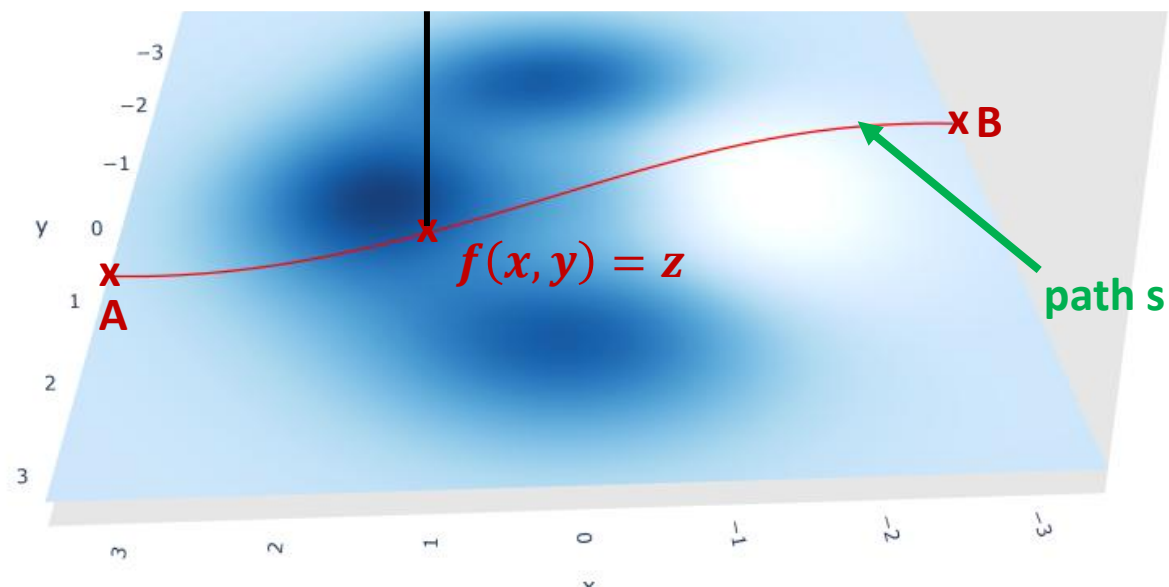
$$\oint_C$$

We can integrate over a
scalar function/field or over a
vector field (see previous slides)

integrating the scalar function
 $f(x, y)$ over the path s

$$\int_A^B f(x, y) ds$$

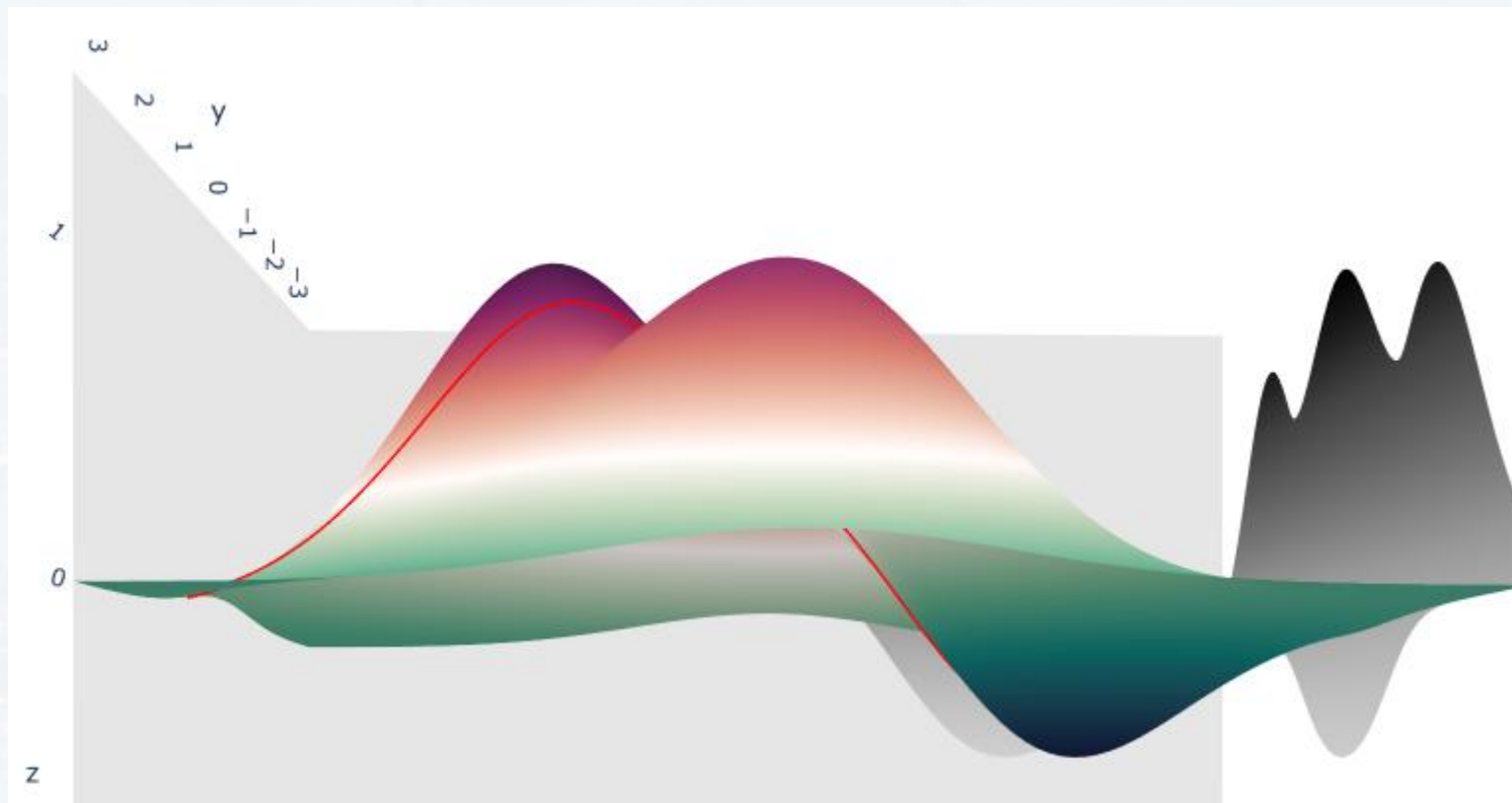
see `PlotLineIntegral.ipynb`





integrating the scalar function $f(x, y)$ over the path s : $\int_A^B f(x, y) ds$

A line integral over a **scalar field** can be interpreted as the area under the curve s along the surface $z = f(x, y)$

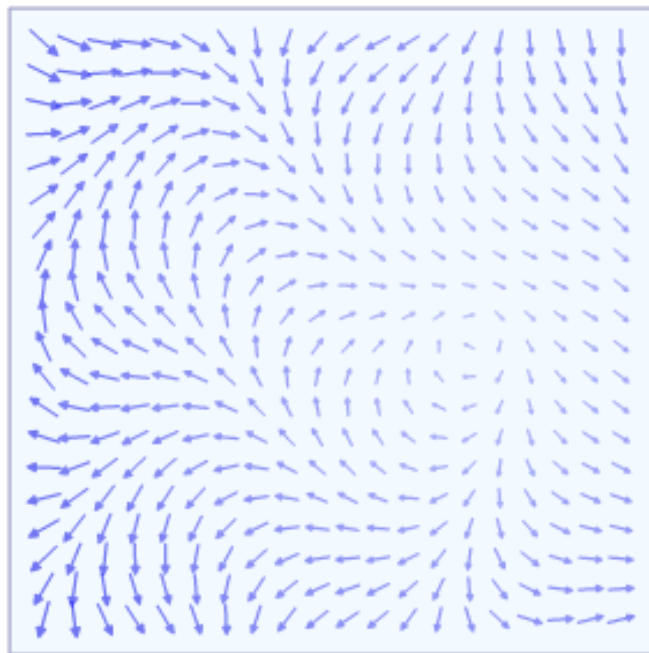


see `PlotLineIntegral.ipynb`



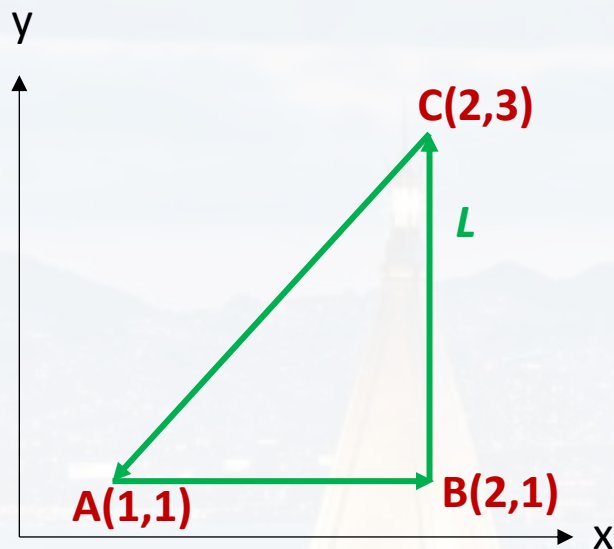
integrating the **vector field** $\vec{v}(x, y)$ over the path s :
$$\int_A^B \vec{v}(x, y) \cdot d\vec{r}$$

A line integral over a **vector field** can be interpreted as multiplying the vector $\vec{v}(x, y)$ with the **tangent vector** $d\vec{r}$ (= current direction) of the path/curve and adding this product up along the curve.





example 1: scalar field $f(x, y) = 2xy^2 + 3$



$$\oint_L f(x, y) ds = \int_A^B f(x, y) ds + \int_B^C f(x, y) ds + \int_C^A f(x, y) ds$$

$$ds^2 = dx^2 + dy^2$$

$$A \rightarrow B \quad y = \text{const} = 1 \quad ds^2 = dx^2 + 0$$

$$\int_A^B f(x, y) ds = \int_{(1,1)}^{(2,1)} 2xy^2 + 3 ds$$

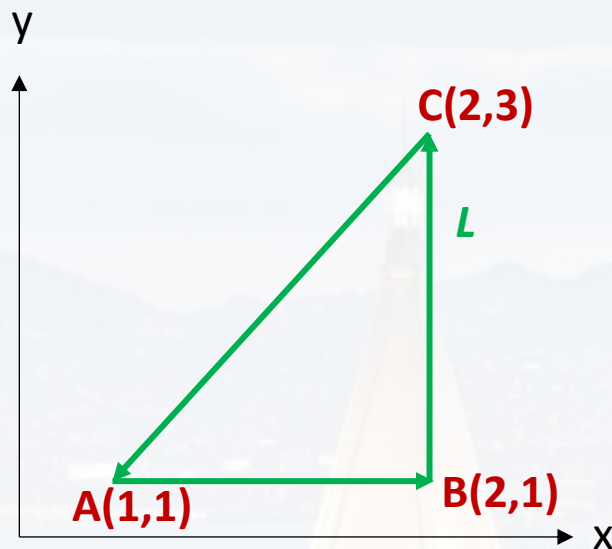
$$= \int_{(1)}^{(2)} 2x dx + \int_{(1)}^{(2)} 3 dx$$

$$= x^2 \Big|_1^2 + 3x \Big|_1^2$$

$$= 4 - 1 + 6 - 3 = 6$$



example 1: scalar field $f(x, y) = 2xy^2 + 3$



$$\oint_L f(x, y) ds = \int_A^B f(x, y) ds + \int_B^C f(x, y) ds + \int_C^A f(x, y) ds$$

$$B \rightarrow C \quad x = \text{const} = 2 \quad ds^2 = 0 + dy^2 \quad ds^2 = dx^2 + dy^2$$

$$\int_B^C f(x, y) ds = \int_{(2,1)}^{(2,3)} 2xy^2 + 3 ds$$

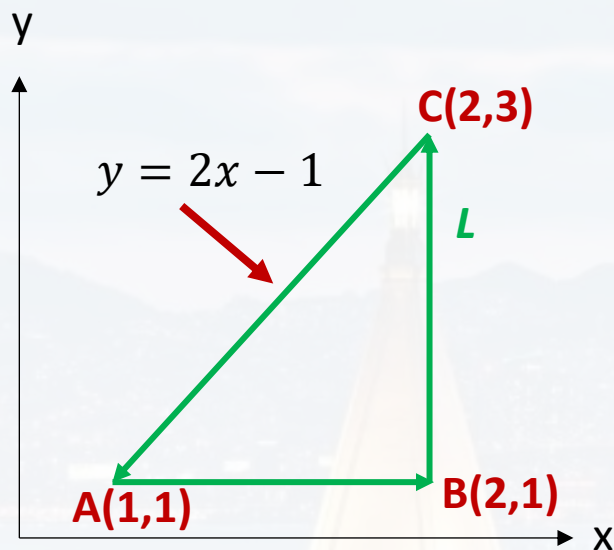
$$= 4 \int_{(1)}^{(3)} y^2 dy + \int_{(1)}^{(3)} 3 dy$$

$$= \frac{4}{3} y^3 \Big|_1^3 + 3y \Big|_1^3$$

$$= 36 - \frac{4}{3} + 9 - 3 = \frac{122}{3}$$



example 1: scalar field $f(x, y) = 2xy^2 + 3$



$$\oint_L f(x, y) ds = \int_A^B f(x, y) ds + \int_B^C f(x, y) ds + \int_C^A f(x, y) ds$$

$$C \rightarrow A \quad y = 2x - 1$$

$$dy = 2dx$$

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{5}dx$$

$$\int_C^A f(x, y) ds = - \int_A^C f(x, y) ds$$

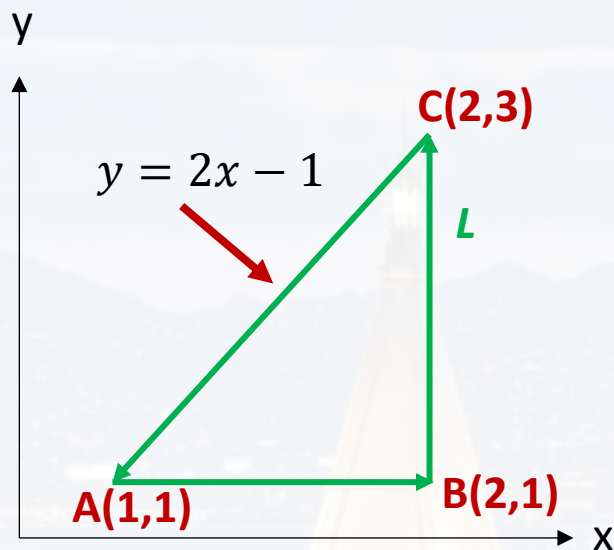
$$= - \int_{(1)}^{(2)} [2x(2x - 1)^2 + 3] \sqrt{5} dx = - \int_{(1)}^{(2)} [2x(4x^2 - 4x + 1) + 3] \sqrt{5} dx$$

$$= -\sqrt{5} \int_{(1)}^{(2)} 8x^3 - 8x^2 + 2x + 3 dx = \sqrt{5} \left[-2x^4 \Big|_1^2 + \frac{8}{3}x^3 \Big|_1^2 - x^2 \Big|_1^2 - 3x \Big|_1^2 \right] = -38.76$$



example 1: scalar field $f(x, y) = 2xy^2 + 3$

We could have also substituted x !



$$\oint_L f(x, y) ds = \int_A^B f(x, y) ds + \int_B^C f(x, y) ds + \int_C^A f(x, y) ds$$

$$C \rightarrow A \quad y = 2x - 1$$

$$\frac{dy}{2} = dx$$

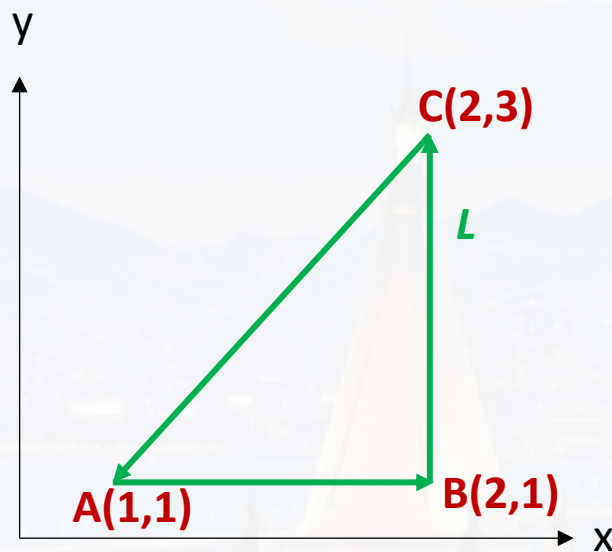
$$ds = \sqrt{dx^2 + dy^2} = \frac{\sqrt{5}}{2} dy$$

$$-\int_A^C f(x, y) ds = -\int_{(1)}^{(3)} \left[2 \left(\frac{y}{2} + \frac{1}{2} \right) y^2 + 3 \right] \frac{\sqrt{5}}{2} dy$$

$$= -\frac{\sqrt{5}}{2} \int_{(1)}^{(3)} y^3 + y^2 + 3 dy = -\frac{\sqrt{5}}{2} \left[\frac{y^4}{4} \Big|_1^3 + \frac{1}{3} y^3 \Big|_1^3 + 3y \Big|_1^3 \right] = -38.76$$



example 1: scalar field $f(x, y) = 2xy^2 + 3$

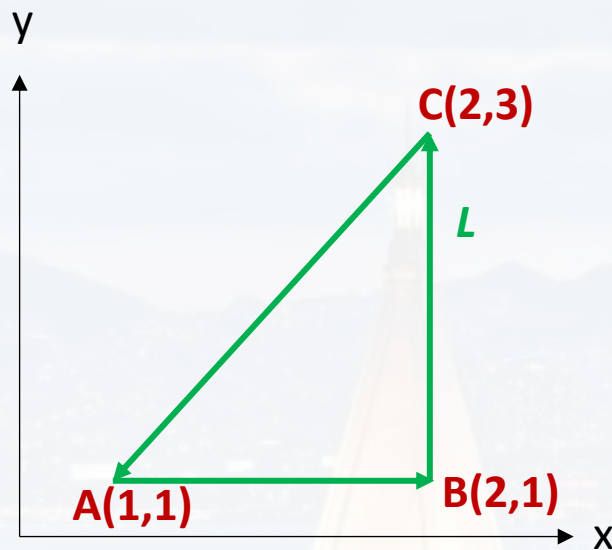


$$\oint_L f(x, y) ds = \int_A^B f(x, y) ds + \int_B^C f(x, y) ds + \int_C^A f(x, y) ds$$

$$= 6 + \frac{122}{3} - 38.76$$



example 2: vector field $\vec{v} = \begin{pmatrix} 2xy^2 \\ 3 \end{pmatrix}$



$$\oint_L \vec{v} d\vec{r} = \int_A^B \vec{v} d\vec{r} + \int_B^C \vec{v} d\vec{r} + \int_C^A \vec{v} d\vec{r}$$

$$A \rightarrow B \quad y = \text{const} = 1$$

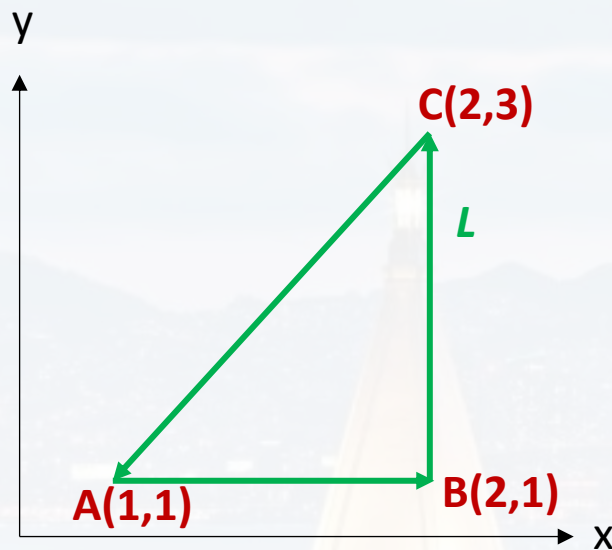
$$\int_A^B \begin{pmatrix} 2xy^2 \\ 3 \end{pmatrix} \begin{pmatrix} dx \\ 0 \end{pmatrix} = \int_1^2 2x dx = x^2 \Big|_1^2 = 3$$

$$B \rightarrow C \quad x = \text{const} = 2$$

$$\int_B^C \begin{pmatrix} 2xy^2 \\ 3 \end{pmatrix} \begin{pmatrix} 0 \\ dy \end{pmatrix} = \int_1^3 3 dy = 3y \Big|_1^3 = 6$$



example 2: vector field $\vec{v} = \begin{pmatrix} 2xy^2 \\ 3 \end{pmatrix}$



$$\oint_L \vec{v} d\vec{r} = \int_A^B \vec{v} d\vec{r} + \int_B^C \vec{v} d\vec{r} + \int_C^A \vec{v} d\vec{r}$$

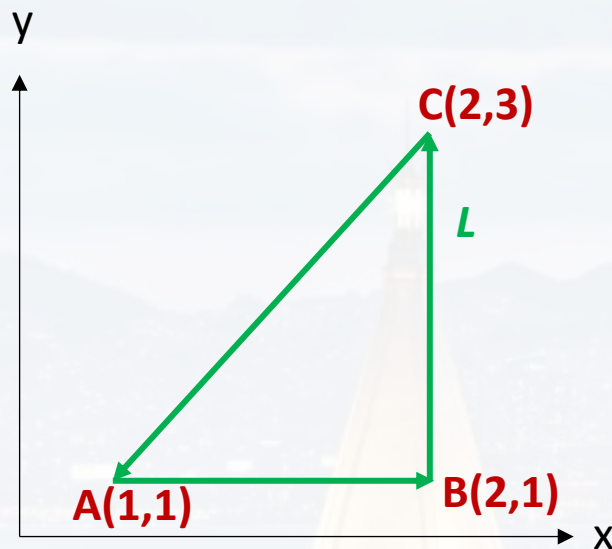
$$C \rightarrow A \quad y = 2x - 1 \quad dy = 2dx$$

$$- \int_A^C \left(\frac{8x^3 - 8x^2 + 2x}{3} \right) \begin{pmatrix} dx \\ 2dx \end{pmatrix} = - \int_1^2 (8x^3 - 8x^2 + 2x + 6) dx$$

$$= -2x^4 \Big|_1^2 + \frac{8}{3}x^3 \Big|_1^2 - x^2 \Big|_1^2 - 6x \Big|_1^2 = -39 + \frac{56}{3} = -20.33$$



example 2: vector field $\vec{v} = \begin{pmatrix} 2xy^2 \\ 3 \end{pmatrix}$



$$\oint_L \vec{v} d\vec{r} = \int_A^B \vec{v} d\vec{r} + \int_B^C \vec{v} d\vec{r} + \int_C^A \vec{v} d\vec{r}$$

$$C \rightarrow A \quad y = 2x - 1 \quad dy = 2dx$$

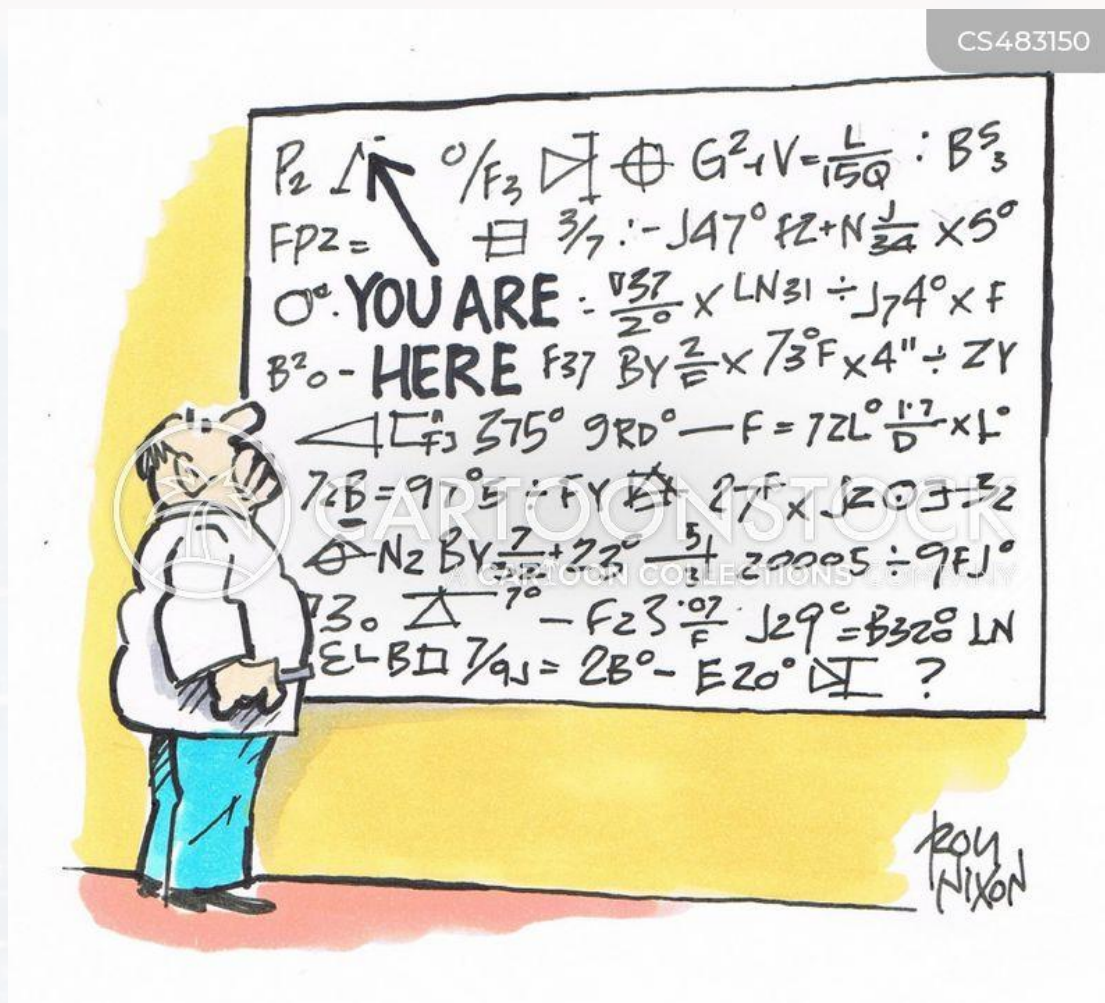
$$- \int_A^C \left(\frac{(y+1)y^2}{3} \right) \left(0.5 \frac{dy}{dy} \right) = - \frac{1}{2} \int_1^3 y^3 + y^2 + 6 dy$$

$$= - \frac{1}{8} y^4 \Big|_1^3 - \frac{1}{6} y^3 \Big|_1^3 - 3y \Big|_1^3 = -20.33$$

$$\text{total: } 3 + 6 - 20.33 = -11.33$$



CS483150



Outline

- Recap: Calculus
- Gradient
- Line Integrals
- **Divergence**
- Curl

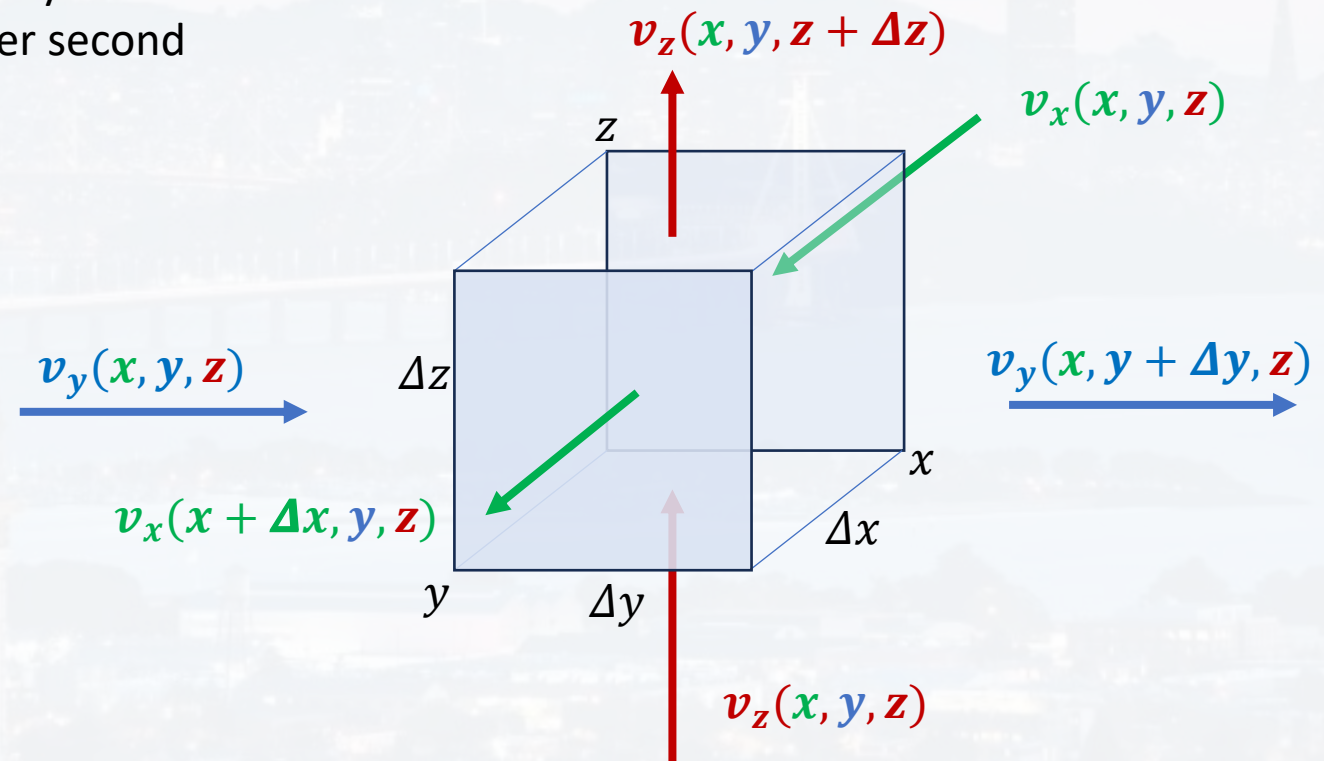


calculating volume flux density

flux: $\Psi = \text{something}/\text{time}$ $\rightarrow$ number/s, mass/s, energy/s etc

flux density: $\varphi = \frac{\text{something}}{\text{time} * \text{area}}$ $\rightarrow$ number/s/m², mass/s/m², energy/s/m² etc

vector field $\vec{v}$, e. g. wind velocity
 $\rightarrow$ flux would be molecules per second





calculating volume flux density

vector field $\vec{v}$, e. g. wind velocity
→ flux would be molecules per second

flux:

$$\Psi = \text{something}/\text{time}$$

flux density:

$$\varphi = \frac{\text{something}}{\text{time} * \text{area}}$$

net flux in **x** direction:

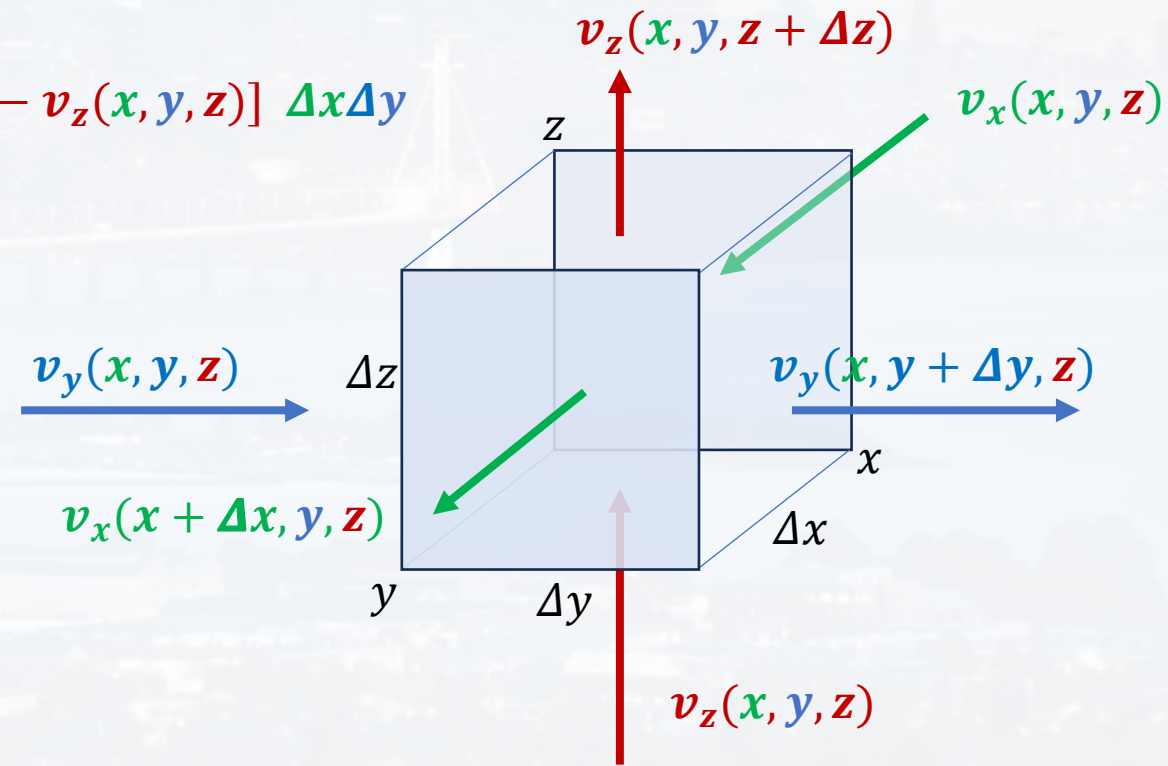
$$[v_x(x + \Delta x, y, z) - v_x(x, y, z)] \Delta y \Delta z$$

net flux in **y** direction:

$$[v_y(x, y + \Delta y, z) - v_y(x, y, z)] \Delta x \Delta z$$

net flux in **z** direction:

$$[v_z(x, y, z + \Delta z) - v_z(x, y, z)] \Delta x \Delta y$$





calculating volume flux density

vector field $\vec{v}$, e. g. wind velocity
 $\rightarrow$ flux would be molecules per second

flux:

$$\Psi = \text{something} / \text{time}$$

flux density:

$$\varphi = \frac{\text{something}}{\text{time} * \text{area}}$$

net flux in **x** direction: $[v_x(x + \Delta x, y, z) - v_x(x, y, z)] \Delta y \Delta z$

net flux in **y** direction: $[v_y(x, y + \Delta y, z) - v_y(x, y, z)] \Delta x \Delta z$

net flux in **z** direction: $[v_z(x, y, z + \Delta z) - v_z(x, y, z)] \Delta x \Delta y$

total net flux:
$$[v_x(x + \Delta x, y, z) - v_x(x, y, z)] \Delta y \Delta z +$$

$$[v_y(x, y + \Delta y, z) - v_y(x, y, z)] \Delta x \Delta z +$$

$$[v_z(x, y, z + \Delta z) - v_z(x, y, z)] \Delta x \Delta y$$

total net flux **per volume**:

$$\frac{[v_x(x + \Delta x, y, z) - v_x(x, y, z)] \Delta y \Delta z + [v_y(x, y + \Delta y, z) - v_y(x, y, z)] \Delta x \Delta z + [v_z(x, y, z + \Delta z) - v_z(x, y, z)] \Delta x \Delta y}{\Delta x \Delta y \Delta z}$$



calculating volume flux density

total net flux **per volume**:

$$\frac{[v_x(x+\Delta x, y, z) - v_x(x, y, z)] \Delta y \Delta z + [v_y(x, y+\Delta y, z) - v_y(x, y, z)] \Delta x \Delta z + [v_z(x, y, z+\Delta z) - v_z(x, y, z)] \Delta x \Delta y}{\Delta x \Delta y \Delta z}$$

total net flux **per volume**:

$$\boxed{\frac{v_x(x+\Delta x, y, z) - v_x(x, y, z)}{\Delta x} + \frac{v_y(x, y+\Delta y, z) - v_y(x, y, z)}{\Delta y} + \frac{v_z(x, y, z+\Delta z) - v_z(x, y, z)}{\Delta z}}$$

first derivatives for $\Delta x, \Delta y, \Delta z \rightarrow 0$

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \equiv \vec{\nabla} \cdot \vec{v} \quad \text{divergence of } \vec{v}$$



calculating volume flux density

divergence

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \equiv \vec{\nabla} \cdot \vec{v} \equiv \mathbf{div} \vec{v}$$

dot product with a
vector ($\vec{v}$),
returns a **scalar**

gradient

$$\left(\frac{\partial}{\partial x} \vec{e}_x + \frac{\partial}{\partial y} \vec{e}_y + \frac{\partial}{\partial z} \vec{e}_z \right) f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix} = \text{grad}(f) \equiv \vec{\nabla} f$$

turns a **scalar** (f)
into a **vector** $\text{grad}(f)$,



$$\vec{\nabla} \cdot \vec{v} \equiv \text{div } \vec{v}$$

divergence: net flux at a given point, if $>0 \rightarrow$ source term
if $<0 \rightarrow$ sink term

$$\frac{[v_x(x+\Delta x, y, z) - v_x(x, y, z)] \Delta y \Delta z + [v_y(x, y+\Delta y, z) - v_y(x, y, z)] \Delta x \Delta z + [v_z(x, y, z+\Delta z) - v_z(x, y, z)] \Delta x \Delta y}{\Delta x \Delta y \Delta z}$$

We derived the divergence, by

- 1) multiplying the vector $\vec{v}$ with the surface element $\Delta x_i \Delta x_j$
- 2) summing this product over all surface elements
- 3) dividing by the volume
- 4) let $\Delta x, \Delta y, \Delta z \rightarrow 0$, i. e. $\partial x, \partial y, \partial z$

$$\text{div } \vec{v} \equiv \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f}$$

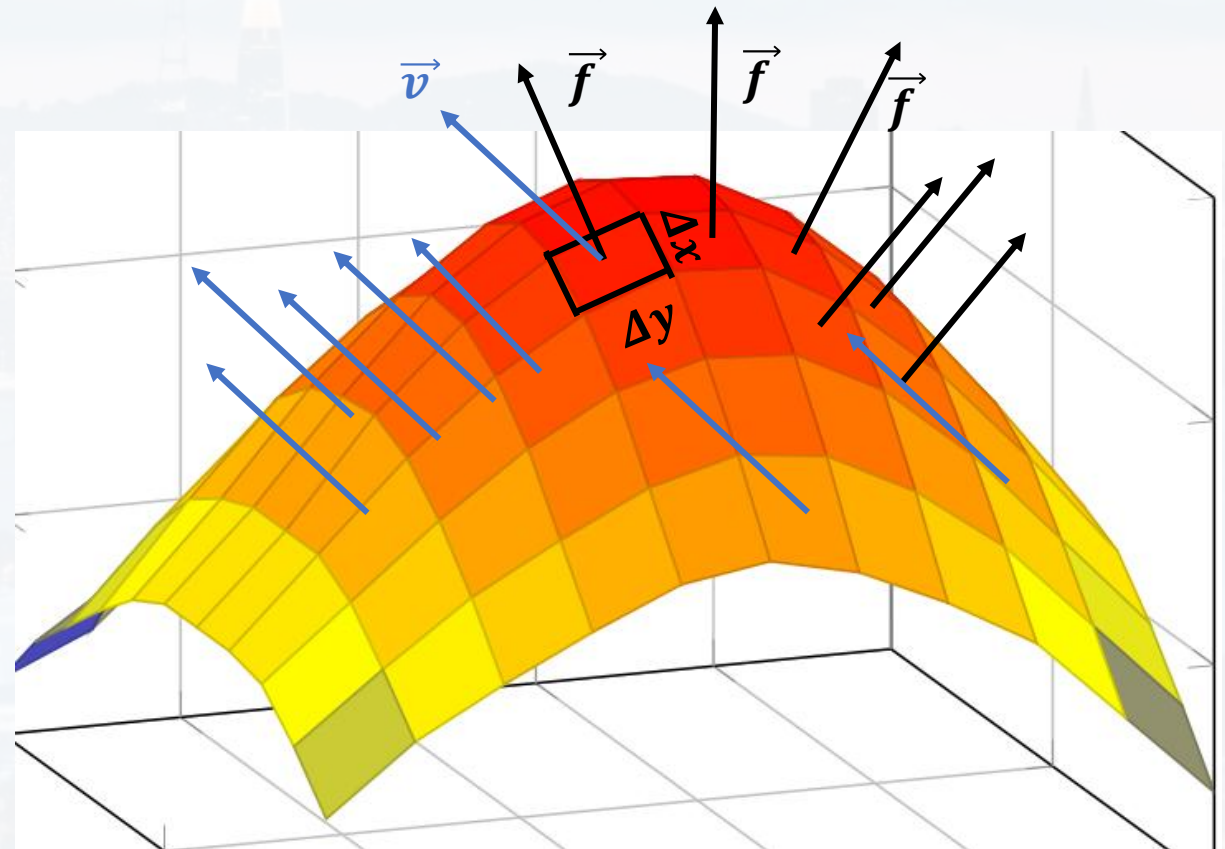


We derived the divergence, by

- 1) multiplying the vector $\vec{v}$ with the surface element $\Delta x_i \Delta x_j$
- 2) summing this product over all surface elements
- 3) dividing by the volume
- 4) let $\Delta x, \Delta y, \Delta z \rightarrow 0$, i. e. $\partial x, \partial y, \partial z$

$$\text{div } \vec{v} \equiv \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f}$$

$\vec{f}$ is **perpendicular** to the surface element





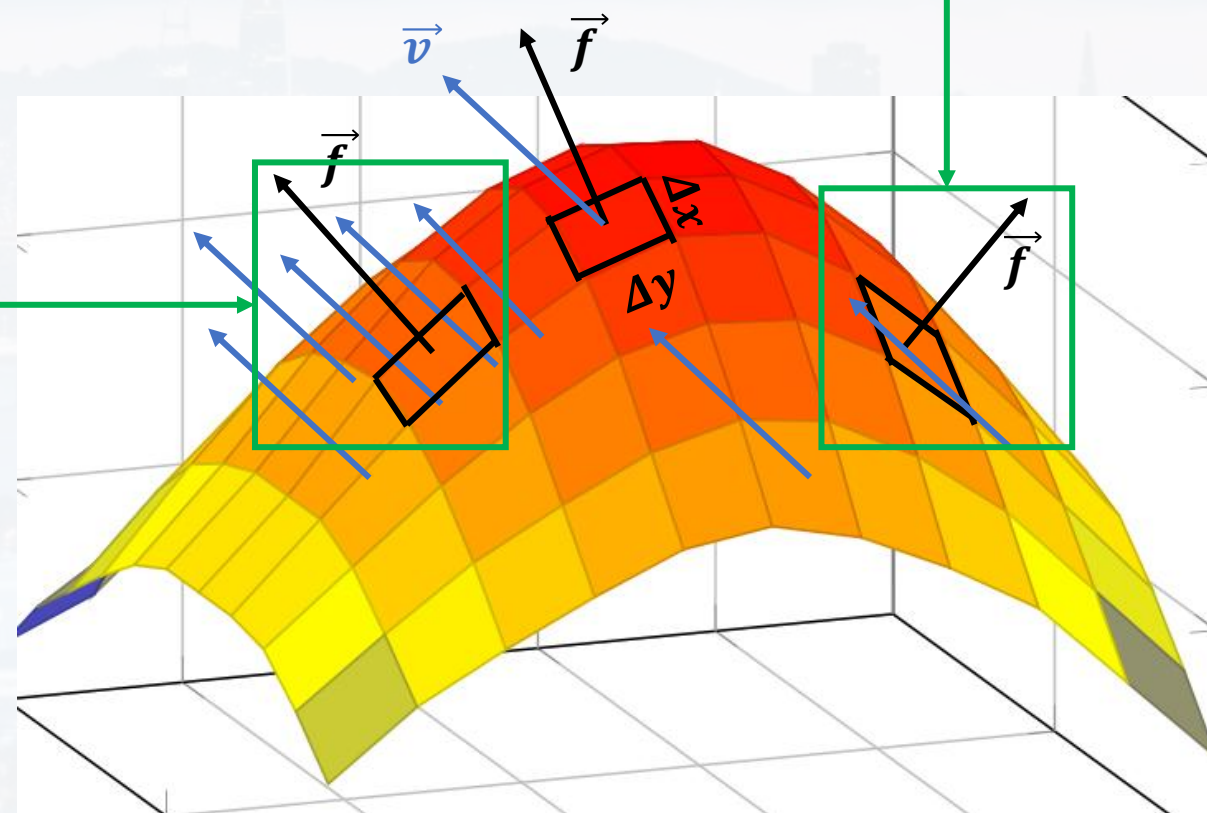
$$\operatorname{div} \vec{v} \equiv \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f} \quad \text{dot product of } \vec{f} \text{ and } \vec{v}$$

no flux in or out:

$$\vec{v} \cdot d\vec{f} = 0$$

max flux out:

$$\vec{v} \cdot d\vec{f} = |\vec{v}|$$





$$\mathbf{div} \vec{v} \equiv \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f}$$

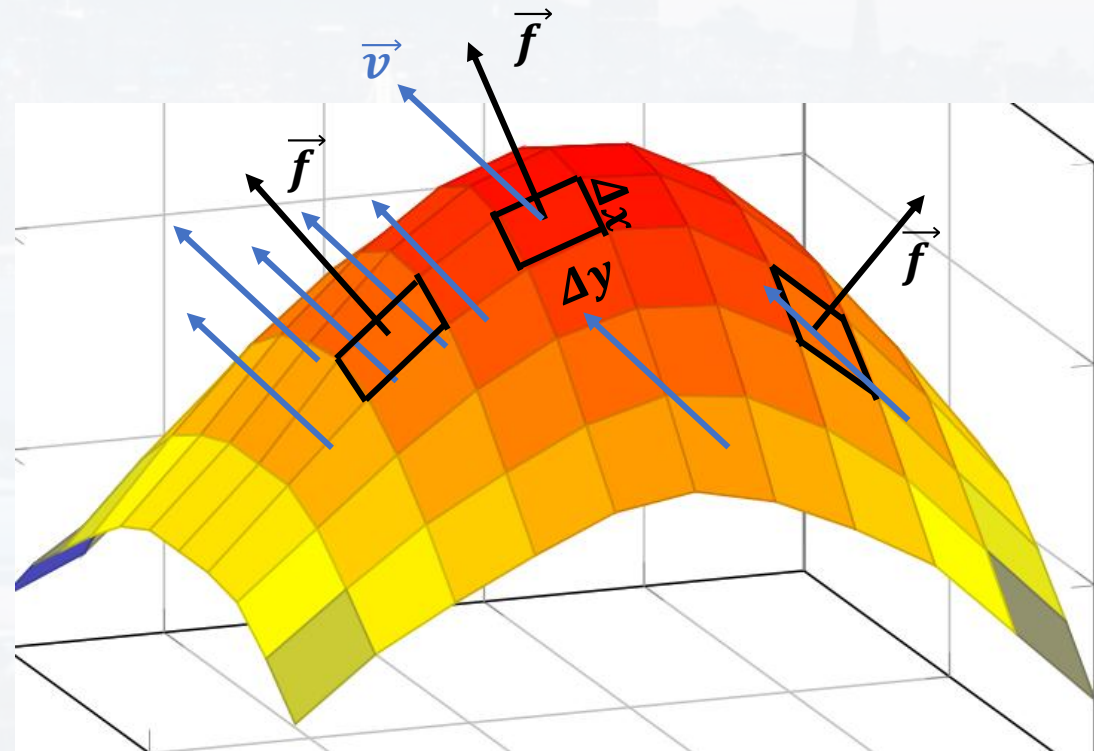
divergence = net flux at a given **point**

summing over the entire volume:

$$\begin{aligned} \int_V \mathbf{div} \vec{v} \cdot dV &= \int_V \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f} \cdot dV \\ &= \int_{(V)} \vec{v} \cdot d\vec{f} \end{aligned}$$

$$\int_V \mathbf{div} \vec{v} \cdot dV = \int_{(V)} \vec{v} \cdot d\vec{f}$$

Gauss' Theorem





$$\mathbf{div} \vec{v} \equiv \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f}$$

divergence = net flux at a given **point**

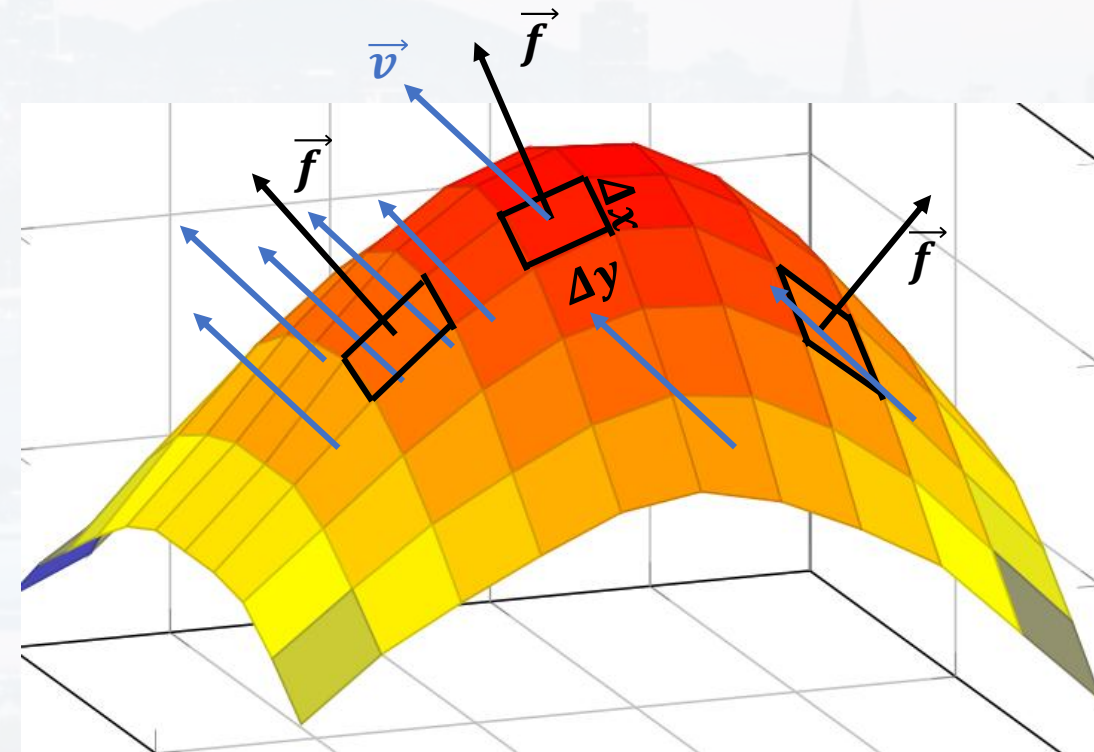
$$\int_V \mathbf{div} \vec{v} \cdot dV = \int_{(V)} \vec{v} \cdot d\vec{f}$$

Gauss' Theorem

Interpretation:

(right-hand side of the equation)
Summing up the flux that goes through all
the surface elements around a given volume,

(left-hand side)
tells us how strong the source/sink within that
volume is.





$$\mathbf{div} \vec{v} \equiv \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f}$$

divergence = net flux at a given **point**

$$\int_V \mathbf{div} \vec{v} \cdot dV = \int_{(V)} \vec{v} \cdot d\vec{f}$$

Gauss' Theorem

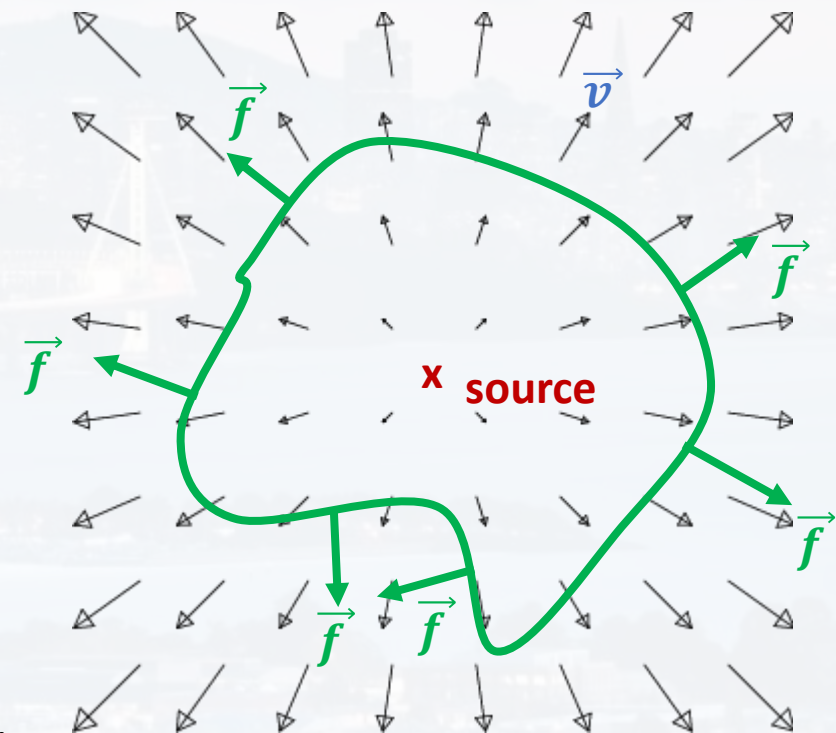
Interpretation:

(right-hand side of the equation)

Summing up the flux that goes through all the surface elements around a given volume,

(left-hand side)

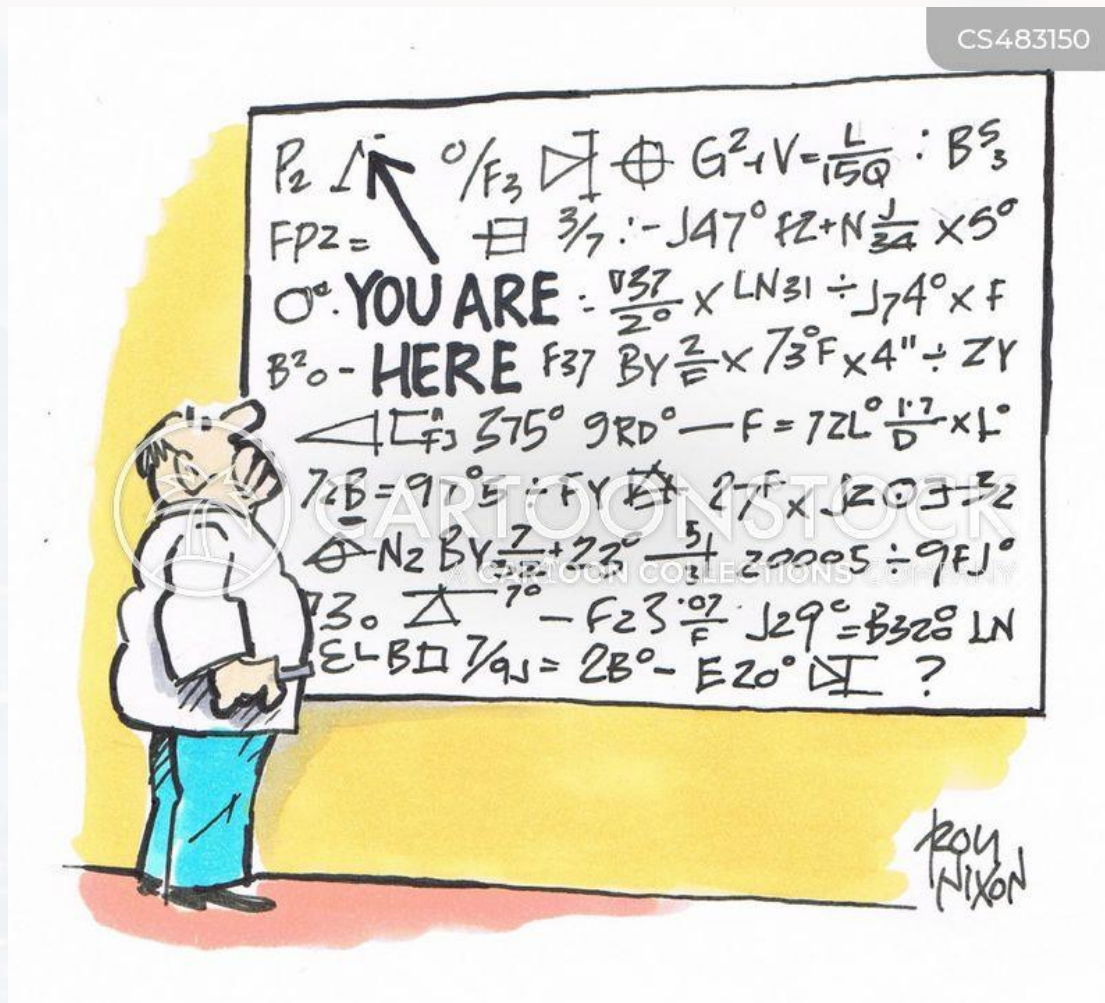
tells us how strong the source/sink within that volume is.



Credit:
Corinne A. Manogue,
Tevian Dray



CS483150



Outline

- Recap: Calculus
- Gradient
- Line Integrals
- Divergence
- **Curl**

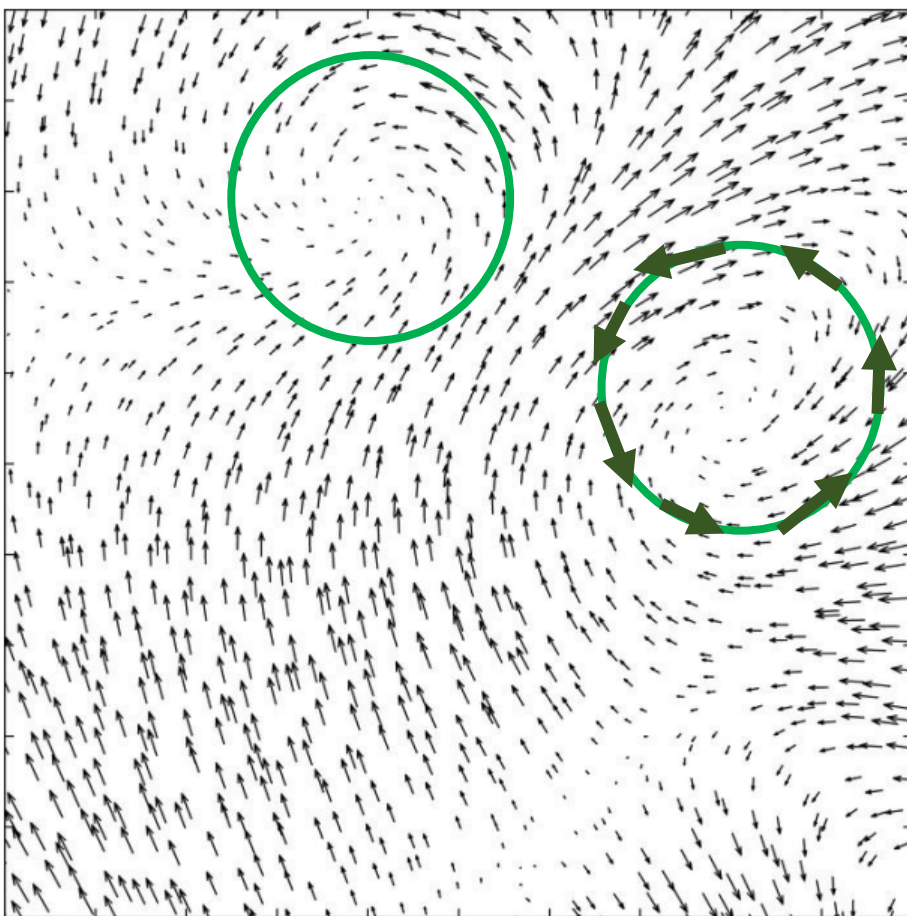


$$\text{div } \vec{v} \equiv \lim_{V \rightarrow 0} \frac{1}{V} \int_{(V)} \vec{v} \cdot d\vec{f}$$

divergence = net flux at a given **point** → tells if source/sink

$\vec{v}$ can also have **curls**

curls



divergence:

in order to get the flux in/out, we had to multiply $d\vec{f}$ with $\vec{v}$

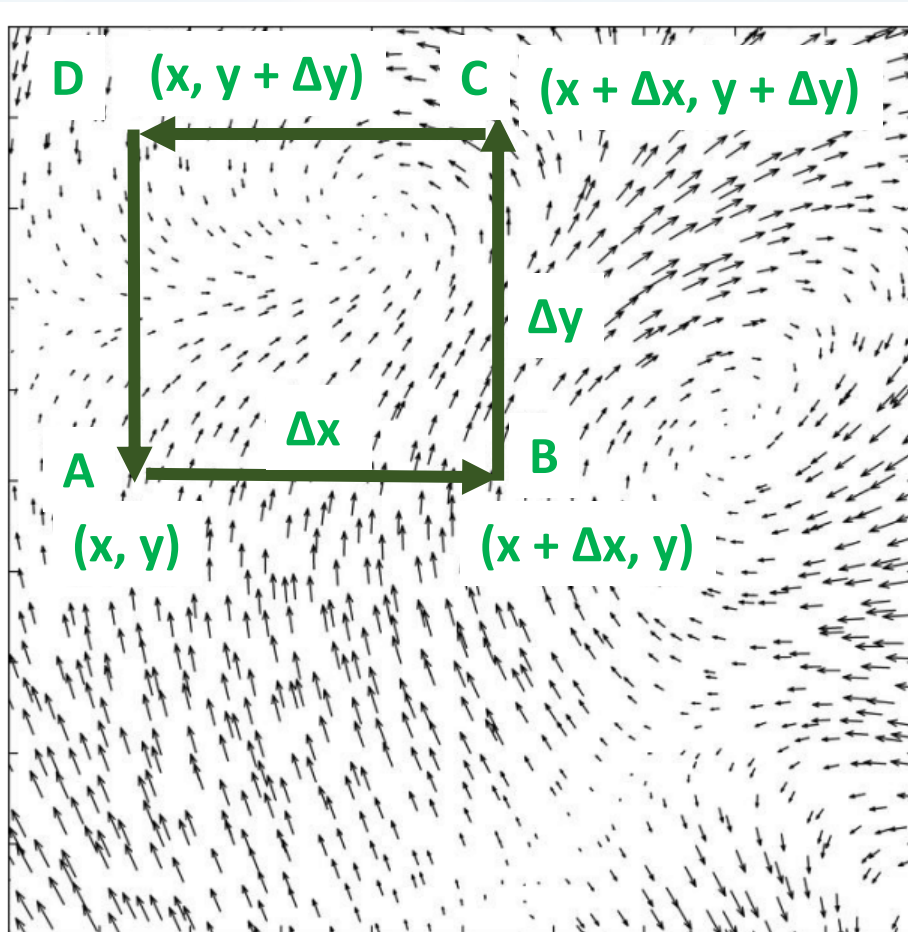
curl:

in order to get the curl/rotation we need to multiply $\vec{v}$ with the **direction $d\vec{r}$ of the path we move along**



curl: in order to get the curl/rotation we need to multiply $\vec{v}$ with the **direction $d\vec{r}$ of the path we move along**

curls



at A) $\Delta\vec{r} = \begin{pmatrix} \Delta x \\ 0 \end{pmatrix}$ $\vec{v} = \begin{pmatrix} v_x(x, y) \\ v_y(x, y) \end{pmatrix}$

at B) $\Delta\vec{r} = \begin{pmatrix} 0 \\ \Delta y \end{pmatrix}$ $\vec{v} = \begin{pmatrix} v_x(x + \Delta x, y) \\ v_y(x + \Delta x, y) \end{pmatrix}$

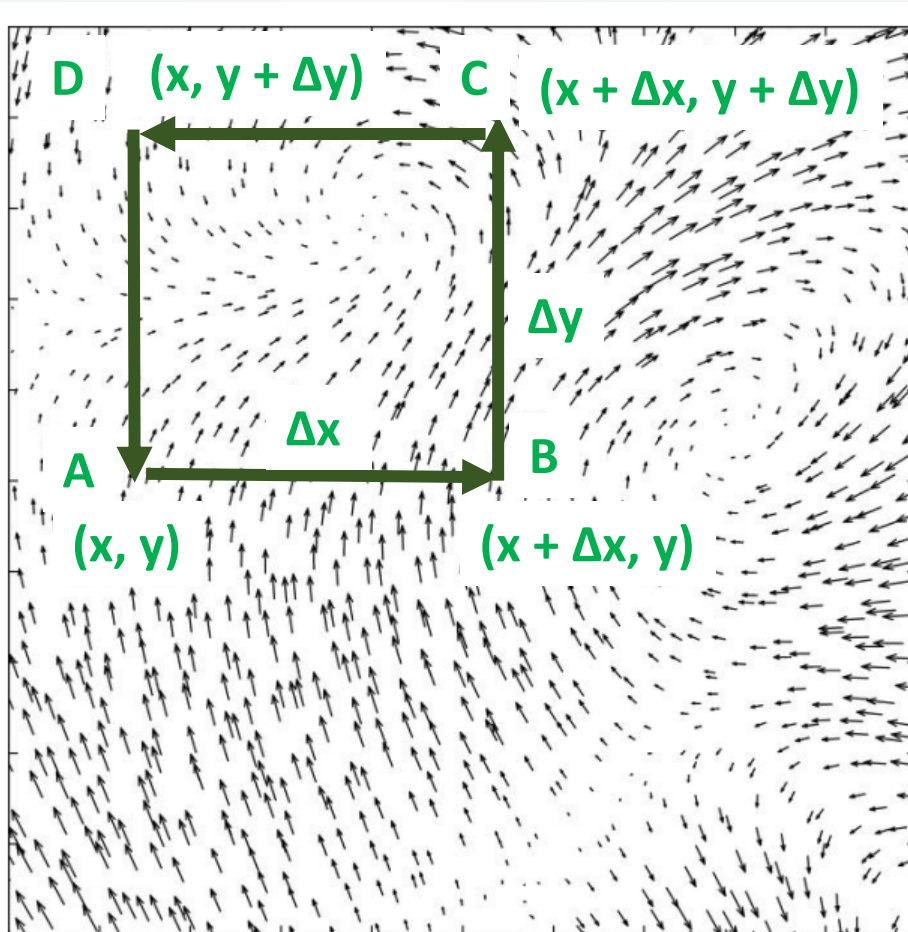
at C) $\Delta\vec{r} = \begin{pmatrix} -\Delta x \\ 0 \end{pmatrix}$ $\vec{v} = \begin{pmatrix} v_x(x + \Delta x, y + \Delta y) \\ v_y(x + \Delta x, y + \Delta y) \end{pmatrix}$

at D) $\Delta\vec{r} = \begin{pmatrix} 0 \\ -\Delta y \end{pmatrix}$ $\vec{v} = \begin{pmatrix} v_x(x, y + \Delta y) \\ v_y(x, y + \Delta y) \end{pmatrix}$



curl: in order to get the curl/rotation we need to multiply $\vec{v}$ with the **direction $d\vec{r}$ of the path we move along**

curls



at A) $\Delta \vec{r} \cdot \vec{v} = v_x(x, y) \Delta x$

at B) $\Delta \vec{r} \cdot \vec{v} = v_y(x + \Delta x, y) \Delta y$

at C) $\Delta \vec{r} \cdot \vec{v} = -v_x(x + \Delta x, y + \Delta y) \Delta x$
 $= -v_x(x, y + \Delta y) \Delta x$

at D) $\Delta \vec{r} \cdot \vec{v} = -v_y(x, y + \Delta y) \Delta y$
 $= -v_y(x, y) \Delta y$



curl: in order to get the curl/rotation we need to multiply $\vec{v}$ with the **direction $d\vec{r}$ of the path we move along**

summing all up and approximating terms like:

$$v_y(x + \Delta x, y) \approx v_y(x, y) + \frac{\partial v_y(x, y)}{\partial x} \Delta x \quad (\text{Taylor series})$$

$$C = \underbrace{v_x(x, y)\Delta x}_{\text{A) to B)} + \underbrace{\left[v_y(x, y) + \frac{\partial v_y(x, y)}{\partial x} \Delta x \right] \Delta y}_{\text{B) to C)} - \underbrace{\left[v_x(x, y) + \frac{\partial v_x(x, y)}{\partial y} \Delta y \right] \Delta x}_{\text{C) to D)} - \underbrace{v_y(x, y)\Delta y}_{\text{D) to A)}}$$

$$C = \frac{\partial v_y(x, y)}{\partial x} \Delta x \Delta y - \frac{\partial v_x(x, y)}{\partial y} \Delta y \Delta x = \underbrace{\left[\frac{\partial v_y(x, y)}{\partial x} - \frac{\partial v_x(x, y)}{\partial y} \right] \Delta y \Delta x}_{\text{curl of } \vec{v}}$$



curl: in order to get the curl/rotation we need to multiply $\vec{v}$ with the **direction $d\vec{r}$ of the path we move along**

$$c = \frac{\partial v_y(x, y)}{\partial x} \Delta x \Delta y - \frac{\partial v_x(x, y)}{\partial y} \Delta y \Delta x = \underbrace{\left[\frac{\partial v_y(x, y)}{\partial x} - \frac{\partial v_x(x, y)}{\partial y} \right]}_{\text{curl of } \vec{v}} \Delta y \Delta x$$

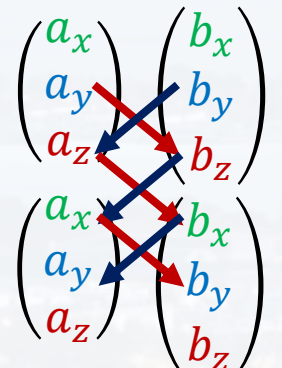
that was 2D, but in 3D:

"curl of $\vec{v}$ " $\equiv rot \vec{v} \equiv \vec{\nabla} \times \vec{v}$

cross product $\vec{a} \times \vec{b}$



trick for 2D and 3D

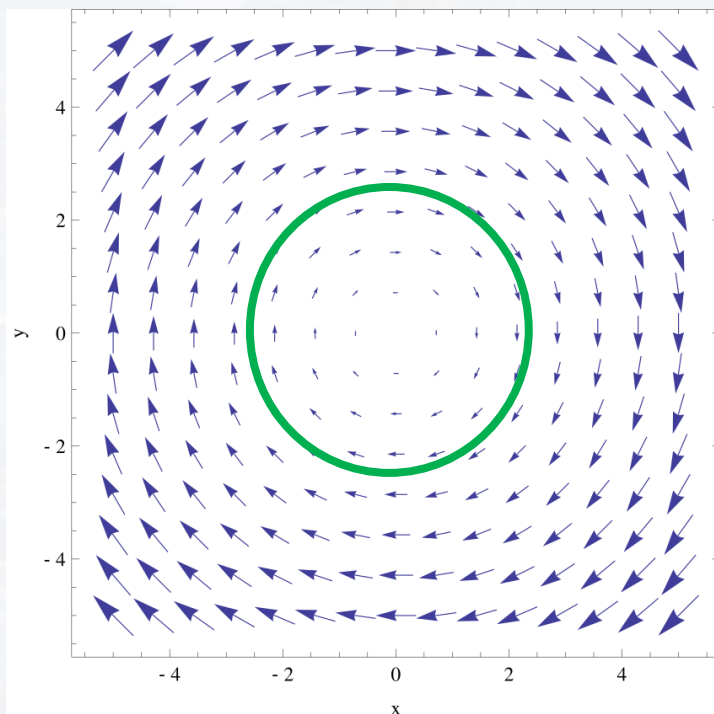
$$\vec{a} \times \vec{b} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} \times \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix} = \begin{pmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{pmatrix}$$




curl: in order to get the curl/rotation we need to multiply $\vec{v}$ with the **direction $d\vec{r}$ of the path we move along**

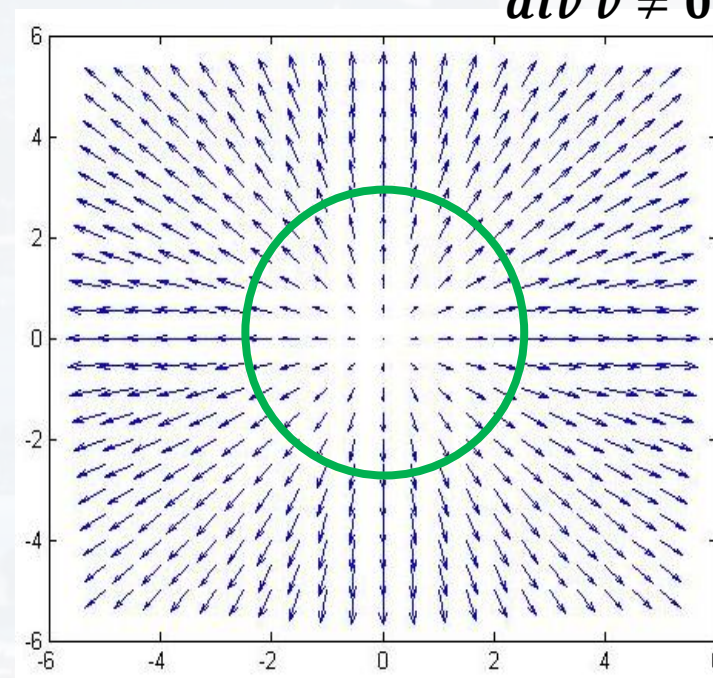
$$\text{"curl of } \vec{v} \text{"} \equiv \text{rot } \vec{v} \equiv \vec{\nabla} \times \vec{v} = \vec{e}_x \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \vec{e}_y \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \vec{e}_z \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

$$\text{rot } \vec{v} \neq 0$$
$$\text{div } \vec{v} = 0$$



credit:
Wikipedia

$$\text{rot } \vec{v} = 0$$
$$\text{div } \vec{v} \neq 0$$



credit:
Stackoverflow



curl $\vec{e}_x \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \vec{e}_y \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \vec{e}_z \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) = \text{rot } \vec{v} \equiv \vec{\nabla} \times \vec{v}$

cross product of two **vectors** ($\vec{\nabla}$ and $\vec{v}$),
returns a **vector**

divergence $\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \equiv \vec{\nabla} \cdot \vec{v} \equiv \text{div } \vec{v}$

dot product with a **vector** ($\vec{v}$),
returns a **scalar**

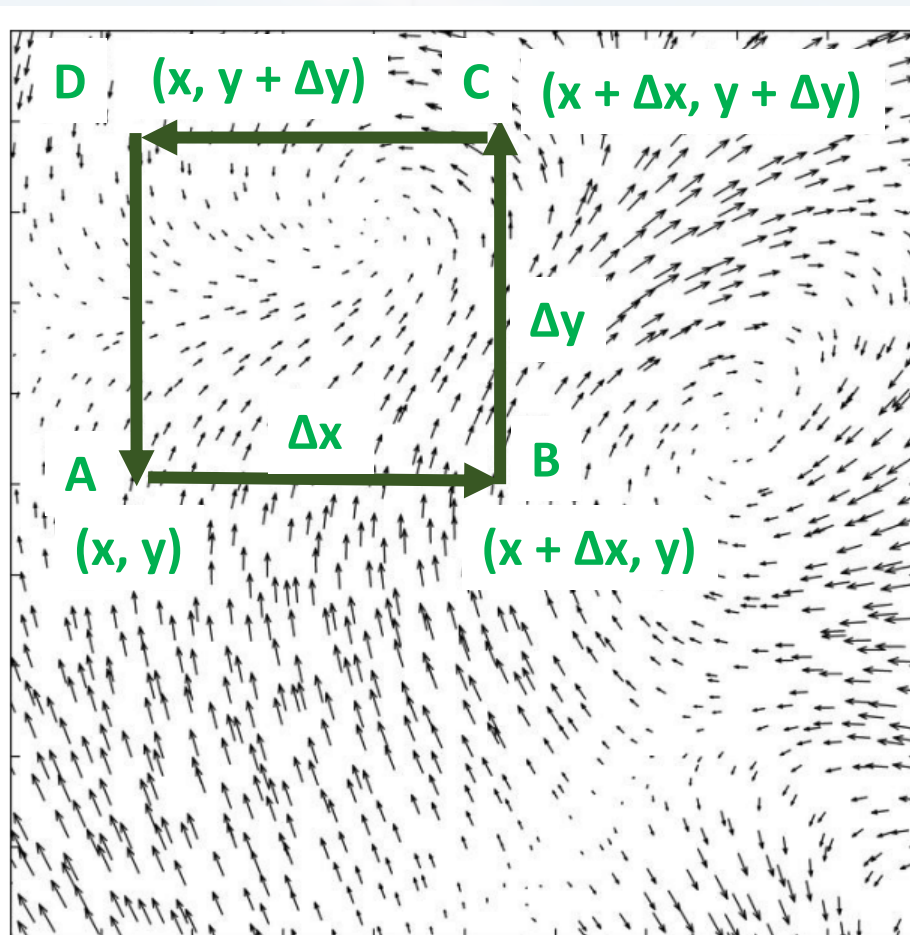
gradient $\left(\frac{\partial}{\partial x} \vec{e}_x + \frac{\partial}{\partial y} \vec{e}_y + \frac{\partial}{\partial z} \vec{e}_z \right) f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix} = \text{grad}(f) \equiv \vec{\nabla} f$

turns a **scalar** (f)
into a **vector** $\text{grad}(f)$,



multiply $\vec{v}$ with the **direction $d\vec{r}$** of the **closed path C we move along** and **adding it all up**

$v_x(x, y)\Delta x$	$+ \left[v_y(x, y) + \frac{\partial v_y(x, y)}{\partial x} \Delta x \right] \Delta y$	$- \left[v_x(x, y) + \frac{\partial v_x(x, y)}{\partial y} \Delta y \right] \Delta x$	$- v_y(x, y)\Delta y =$
A) to B)	B) to C)	C) to D)	D) to A)



equals the **difference** of the **mixed partial derivatives**, **times the area A defined by C**

$$= \left[\frac{\partial v_y(x, y)}{\partial x} - \frac{\partial v_x(x, y)}{\partial y} \right] \Delta y \Delta x$$

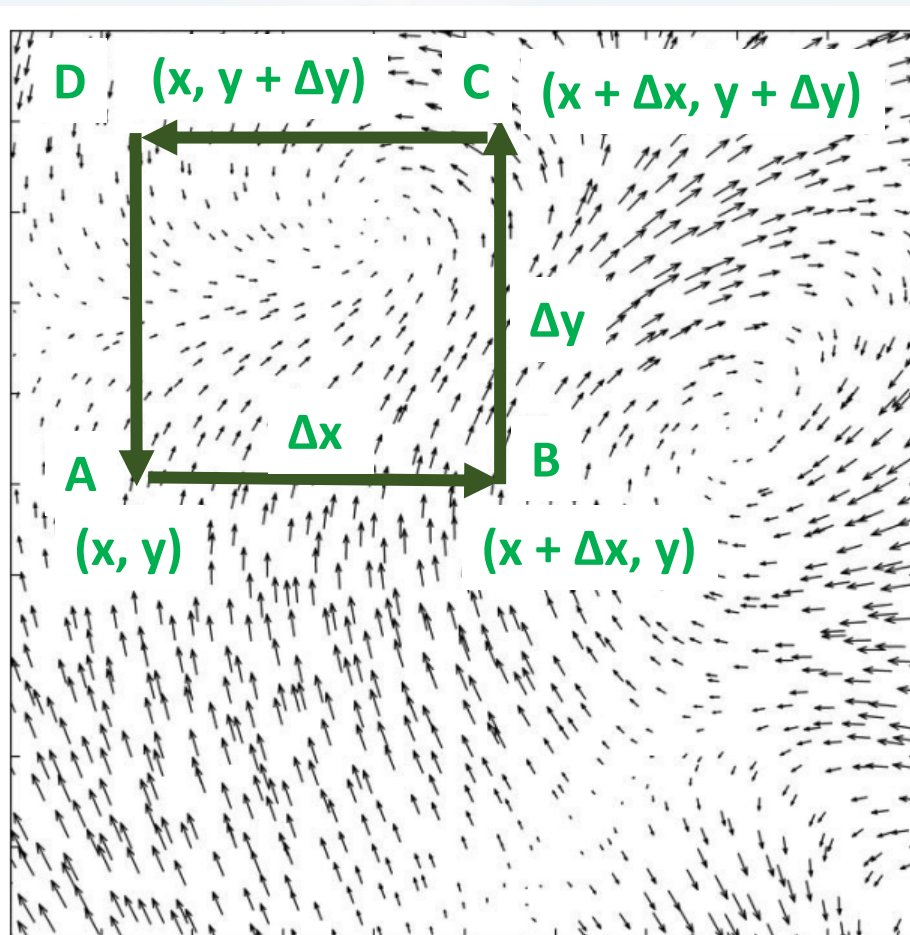
$$\oint_C \vec{v} \cdot d\vec{r} = \iint \left[\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right] dA \quad \text{Green's Theorem (Stoke's Theorem in 2D)}$$

$$\oint_C \vec{v} \cdot d\vec{r} = \int_A (\vec{\nabla} \times \vec{v}) \cdot d\vec{f} \quad \text{Stoke's Theorem}$$



multiply $\vec{v}$ with the **direction** $d\vec{r}$ of the closed path C we move along and adding it all up

$$v_x(x, y)\Delta x + \left[v_y(x, y) + \frac{\partial v_y(x, y)}{\partial x} \Delta x \right] \Delta y - \left[v_x(x, y) + \frac{\partial v_x(x, y)}{\partial y} \Delta y \right] \Delta x - v_y(x, y)\Delta y =$$



equals the difference of the mixed partial derivatives, times the area A defined by C

$$= \left[\frac{\partial v_y(x, y)}{\partial x} - \frac{\partial v_x(x, y)}{\partial y} \right] \Delta y \Delta x$$

$$\oint_C \vec{v} \cdot d\vec{r} = \iint \left[\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right] dA$$

Green's Theorem
(Stoke's Theorem in 2D)

$$\oint_C \vec{v} \cdot d\vec{r} = \int_A (\vec{\nabla} \times \vec{v}) \cdot d\vec{f}$$

Stoke's Theorem



Thank you very much for your attention!

